

Task 04

Continue in the document created in Task 01 and perform the following operations. Print and attach your final result. *No screenshots allowed.*

1. Make a heading of the task using the same formatting as Task 01.
2. Using advanced features of Word, create your CV.
3. Add a watermark only to your CV page.
4. Create the following multilevel list using the same list provided.
 1. Video provides a powerful way to help you prove your point.
 - 1.1. When you click Online Video, you can paste in the embed code for the video you want to add.
 - 1.1.1. You can also type a keyword to search online for the video that best fits your document.
 2. To make your document look professionally produced.
 - 2.1. For example, you can add a matching cover page, header, and sidebar.
 - 2.1.1. Click Insert and then choose the elements you want from the different galleries.
 - 2.1.2. Themes and styles also help keep your document coordinated.
 - 2.1.3. When you click Design and choose a new Theme, the pictures, charts.
 3. You can also type a keyword to search online for the video that best fits your document.
 - 3.1. To make your document look professionally produced.
 4. For example, you can add a matching cover page, header, and sidebar.
 - 4.1. Click Insert and then choose the elements you want from the different galleries.
 - 4.2. Themes and styles also help keep your document coordinated.
 - 4.3. When you click Design and choose a new Theme, the pictures, charts.
 - 4.3.1. When you apply styles, your headings change to match the new theme. Save time in Word with new buttons that show up where you need them.
 - 4.3.2. To make your document look professionally produced.
 - 4.3.2.1. For example, you can add a matching cover page, header, and sidebar.
 - 4.3.2.2. Click Insert and then choose the elements you want from the different galleries.
 - 4.3.2.3. Themes and styles also help keep your document coordinated.
 - 4.3.2.3.1. When you click Design and choose a new Theme, the pictures, charts.
 - 4.3.2.3.2. When you apply styles, your headings change to match the new theme.
 - 4.3.2.3.3. Save time in Word with new buttons that show up where you need them. To change the way a picture fits in your document, click it and a button for layout options appears next to it.

SUBMISSION GUIDELINES

- Prepare a single word file labeled with Roll No and Lab No. e.g. 'SE2023xxx_Lab04'.
- Convert the file into PDF.
- Submit the file at gcr and hardcopy
- -100% policies for plagiarism.

NED University of Engineering & Technology
Department of Software Engineering

**Course Code & Title: CT-174 FIT Assessment Rubric
for Psychomotor (P2-PLO-3)**

Instructor: _____

Batch: 2024

Class: FESE

Semester: Fall 2024.

Student's Name: _____

Roll No.: _____

Laboratory Session No. _____

Date: _____

Psychomotor Domain Assessment Rubric-Level P2-PLO3					
Skill Sets	Extent of Achievement				
	Below Average (0--1)	Average (2)	Good (3)	Very Good (4)	Excellent (5)
Problem Definition and Understanding	Unable to define and understand the problem clearly.	Struggles to define the problem clearly, with significant gaps in understanding	Defines the problem, but with gaps or a limited understanding of key components, impacting the overall quality of definition.	Effectively defines the problem with a good understanding of its components, showcasing proficiency in problem definition.	Demonstrates an exceptional understanding of the problem, identifying all key components and challenges with insightful depth.
Technical Implementation <i>Proficient use of tools to complete the problem</i>	Struggles to use tools effectively, requiring significant improvement.	Uses tools with noticeable inefficiencies.	Demonstrates basic proficiency in utilizing tool, with some room for improvement.	Effectively uses tools to support the task.	Integrates use of tools seamlessly to enhance the design and development process
Solution Design& Accuracy	Struggles to relate the implemented solutions to the design and development goals, requiring significant improvement.	Shows limited connection between the implemented solutions and the design and development context.	Demonstrates a basic understanding of how the solutions align with design and development requirements.	Effectively relates the implemented solutions to the overall design and development objectives	Maps the implemented solutions seamlessly to the overarching design and development goals, demonstrating a comprehensive understanding.
Trouble Shooting Skills Issue resolution skills	Unable to trouble shoot	Limited troubleshooting	Adequate troubleshooting, minor errors remaining	Good troubleshooting, no errors remaining	Excellent trouble shooting no errors remain without delay.
Document Organization and on time submission	Poor Unclear disorganized document late submission	Somewhat clear with lack of organization and structuring late submission	Somewhat clear with lack of organization and structuring on time submission	Reasonably clear and organized with adequate structuring on time submission	Highly clear and well-structured document, On time submit

Weighted CLO (Psychomotor Score)		
Instructor's Signature with Date		

Lab Session: 05

OBJECTIVE

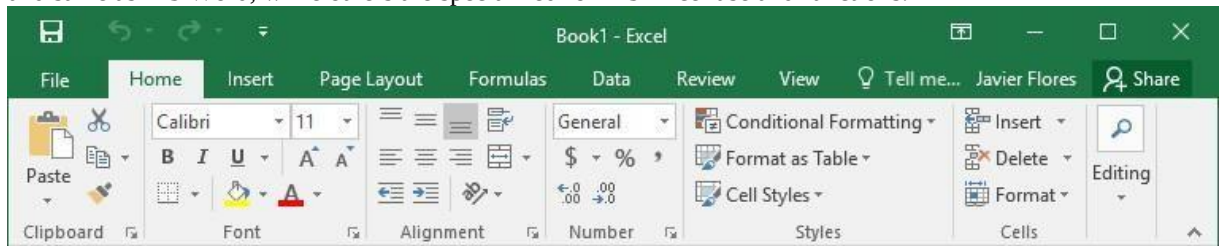
To **practice** with the MS Excel environment

THEORY

Microsoft Excel is a spreadsheet editor developed by Microsoft. It features calculation or computation capabilities, graphing tools, pivot tables, and a macro programming language called Visual Basic for Applications. It allows users to store, organize, and analyze information. Excel forms part of the Microsoft 365 suite of software.

Ribbon

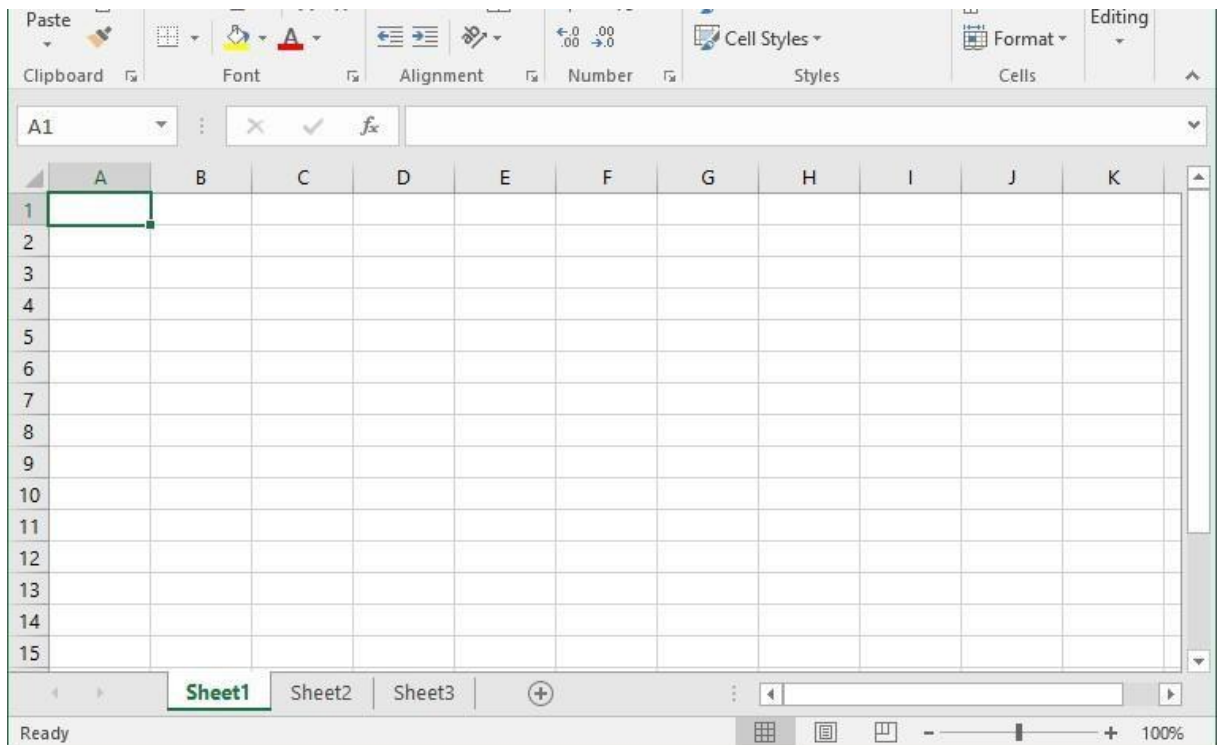
MS Excel follows the same appearance as MS Word by using a Ribbon. Some tabs and menu items on the Ribbon are standard and same as MS Word, while others are specialized for MS Excel use and functions.



Each tab in the ribbon contains several groups of related commands. Some groups also have a small arrow in the bottom-right corner for even more options.

Worksheet

When Excel starts, a blank workbook is open by default. A workbook is just an Excel file.



A workbook contains several worksheets. A worksheet (also known as a spreadsheet, or sheet) consists of cells in which users can enter and calculate data. The cells are organized into columns and rows. The worksheets within an excel file can be switched by clicking on the desired tab at the bottom of the Excel window. Sheets can also be renamed by right-clicking on them and selecting the rename option.

The Quick Access Toolbar

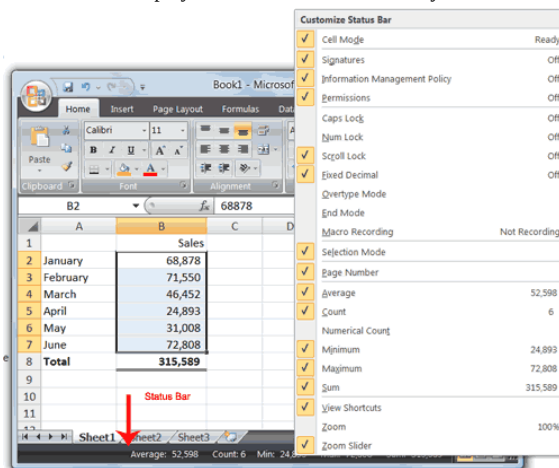


Next to the Microsoft Office button is the Quick Access toolbar. The Quick Access toolbar gives you with access to commands you frequently use. By default, Save, Undo, and Redo appear on the Quick Access toolbar. You can use Save to save your file, Undo to roll back an action you have taken, and Redo to reapply an action you have rolled back.

The Title Bar



Next to the Quick Access toolbar is the Title bar. On the Title bar, Microsoft Excel displays the name of the workbook you are currently



using. At the top of the Excel window, you should see "Microsoft Excel - Book1" or a similar name.

The Formula Bar



Formula Bar

If the Formula bar is turned on, the cell address of the cell you are in displays in the Name box which is located on the left side of the Formula bar. Cell entries display on the right side of the Formula bar. If you do not see the Formula bar in your window, perform the following steps:

1. Choose the View tab.
2. Click Formula Bar in the Show/Hide group. The Formula bar appears.

Note: The current cell address displays on the left side of the Formula bar.

The Status Bar

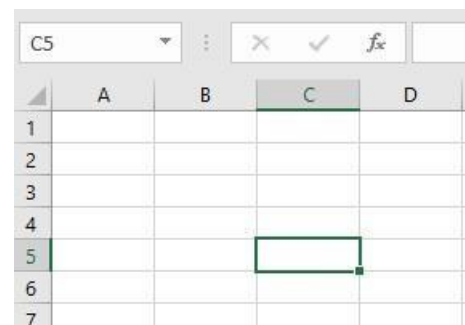
The Status bar appears at the very bottom of the Excel window and provides such information as the sum, average, minimum, and maximum value of selected numbers. You can change what displays on the Status bar by right-clicking on the Status bar and selecting the options you want from the Customize Status Bar menu. You click a menu item to select it. You click it again to deselect it. A check mark next to an item means the item is selected.

Cell

Each rectangle in a worksheet is called a cell. It is the intersection of a row and a column. Cells are the basic building blocks of a worksheet. Every worksheet is made up of thousands of cells.

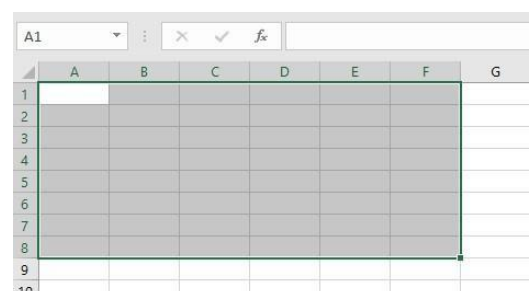
Columns in Excel are labeled using alphabets and run from the top to the bottom of the worksheet. Rows are labeled using numbers and run from the left of the worksheet to the right.

Each cell has a name/identity of itself called the cell address. The name of cell is the "name of column + number of row". The presently selected cell is highlighted by a colored border around it and cell's column and row headings are highlighted when the cell is selected. Its name is also displayed in the name box on the top-left of the worksheet. For example, in the image, the selected cell is C5.



Multiple cells can also be selected simultaneously. A group of cells is called a range. Ranges are referred to using the first and last cell in the cell range. For example, C1:C5 is a range of five consecutive cells in column C, i.e., C1, C2, C3, C4, and C5.

Ranges can also span multiple rows and columns, as shown. In this case, the cell range is A1:F8. Moreover, cell ranges can be named to keep track of important cell ranges in the spreadsheet. This can be done by selecting the cell range, then selecting **Define Name** on the **Formulas** tab.



Cell Content

All information entered into the worksheet is stored in cells. Each cell can contain different types of content, including text (letters, numbers, dates), formatting (formatted text, colored backgrounds, etc.), and formulas and functions that calculate the cells values.

Inserting Content

To insert content in a cell, simply select a cell and insert data from the keyboard or paste previously copied data. The content will appear in the cell and the formula bar.

Deleting Content

Select the cell or range to clear. Then select **Clear Contents** under the **Clear** command on the **Home** tab. Alternatively, press *Del/Delete* key on the keyboard.

Copying/Pasting Content

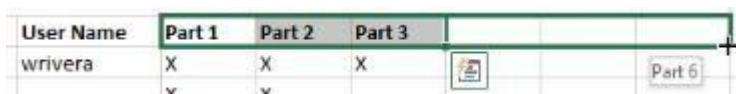
Select the cell/range that will be copied. Right-click on the selection and select **Copy** or press *Ctrl+C* on the keyboard. The copied cell(s) will have a dashed box around them.

Select the cell/range to paste the content. Right-click on the selection and select **Paste** or press *Ctrl+V* on the keyboard.

Fill Handle

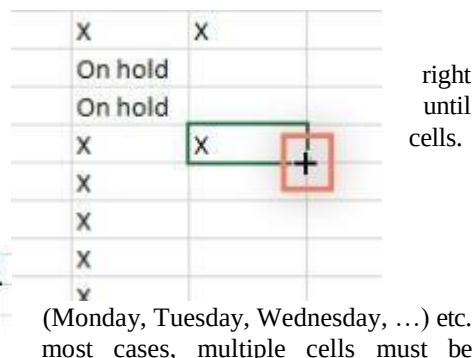
Fill handle is particularly useful when copying content into adjacent cells. Selects the cell(s) that will be copied, then hover the mouse over the lower-corner of the cell so the fill handle appears. Click and drag the fill handle all of the cells to be filled are selected. Release the mouse to fill the selected

Fill handle can also be used to continue a series, i.e., a range of cells that follows a sequential order such as numbers (1, 2, 3, ...) or days of the week



User Name	Part 1	Part 2	Part 3	Part 6
wrivera	X	X	X	y

In selected before using the fill handle to help Excel determine the series order.



To use the fill handle to continue a series, select the cell range that contains the series. Click and drag the fill handle to continue the series. Release the mouse. If Excel understood the series, it will be continued in the selected cells.

Adding Cells

This will insert an entire row/column in the worksheet. Right-click on the row/column where the new cells will be inserted, then select **Insert**.

Deleting Cells

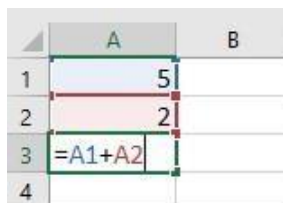
This will delete an entire row/column in the worksheet. Right-click on the row/column that will be deleted, then select **Delete**.

Formulas and Functions

Formulas

One of the most powerful features in Excel is the ability to calculate numerical information using formulas. Just like a calculator, Excel can add, subtract, multiply, and divide, and uses standard operators for formulas.

- + Addition
- Subtraction
- * Multiplication
- / Division
- ^ Exponents



	A	B
1	5	
2	2	
3	=A1+A2	
4		

All formulas in Excel must begin with an equals sign (=). This is because the cell contains, or is equal to, the formula and the value it calculates. Formulas can be created using numbers or cell addresses; the latter is more common. In the example shown, cell A3 will contain the sum of cell A1 and A2. If the content of either A1 and/or A2 is changed, the change will be reflected in cell A3.

Formulas can also include a combination of cell references and numbers, e.g., $N10 \times 1.05$. Formulas can also be copied to adjacent cells with the fill handle. Formulas can also be edited either in the cell or in the formula bar.

Move Around a Worksheet

By using the arrow keys, you can move around your worksheet. You can use the down arrow key to move downward one cell at a time. You can use the up arrow key to move upward one cell at a time. You can use the Tab key to move across

the page to the right, one cell at a time. You can hold down the Shift key and then press the Tab key to move to the left, one cell at a time. You can use the right and left arrow keys to move right or left one cell at a time. The Page Up and Page Down keys move up and down one page at a time. If you hold down the Ctrl key and then press the Home key, you move to the beginning of the worksheet.

Move Around the Worksheet

The Down Arrow Key

- Press the down arrow key several times. Note that the cursor moves downward one cell at a time.

The Up Arrow Key

- Press the up arrow key several times. Note that the cursor moves upward one cell at a time.

The Tab Key

1. Move to cell A1.
2. Press the Tab key several times. Note that the cursor moves to the right one cell at a time.

The Shift+Tab Keys

- Hold down the Shift key and then press Tab. Note that the cursor moves to the left one cell at a time.

The Right and Left Arrow Keys

1. Press the right arrow key several times. Note that the cursor moves to the right.
2. Press the left arrow key several times. Note that the cursor moves to the left.

Page Up and Page Down

1. Press the Page Down key. Note that the cursor moves down one page.
2. Press the Page Up key. Note that the cursor moves up one page.

The Ctrl-Home Key

1. Move the cursor to column J.
2. Stay in column J and move the cursor to row 20.

Hold down the Ctrl key while you press the Home

1. key. Excel moves to cell A1.

Go To Cells Quickly

The following are shortcuts for moving quickly from one cell in a worksheet to a cell in a different part of the worksheet.

Go to -- F5

The F5 function key is the "Go To" key. If you press the F5 key, you are prompted for the cell to which you wish to go. Enter the cell address, and the cursor jumps to that cell.

1. Press F5. The Go To dialog box opens.
2. Type **J3** in the Reference field.

The Name Box

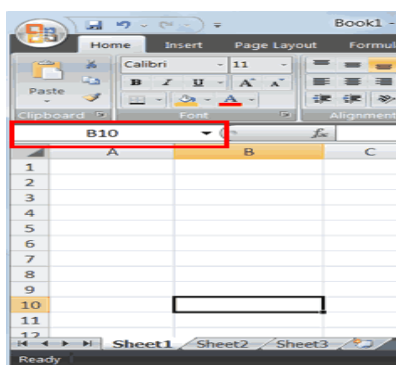
You can also use the Name box to go to a specific cell. Just type the cell you want to go to in the Name box and then press Enter.

3. Press Enter. Excel moves to cell J3.

Go to -- Ctrl+G

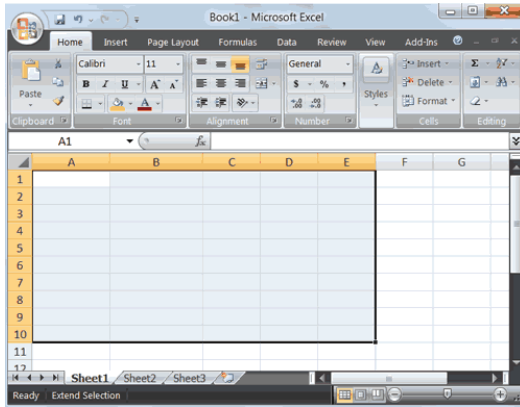
You can also use Ctrl+G to go to a specific cell.

1. Hold down the Ctrl key while you press "g" (Ctrl+g). The Go To dialog box opens.
2. Type **C4** in the Reference field.
3. Press Enter. Excel moves to cell C4.



1. Type **B10** in the Name box.
2. Press Enter. Excel moves to cell B10.

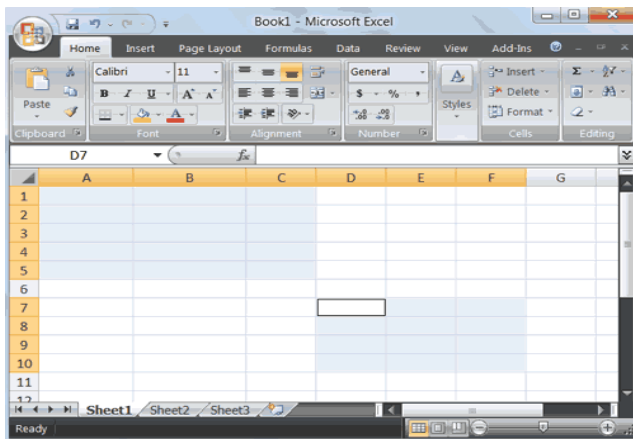
Select Cells



If you wish to perform a function on a group of cells, you must first select those cells by highlighting them. The exercises that follow teach you how to select.

Alternative Method: Select Cells by Dragging

You can also select an area by holding down the left mouse button and dragging the mouse over the area. In addition, you can select noncontiguous areas of the worksheet by doing the following:



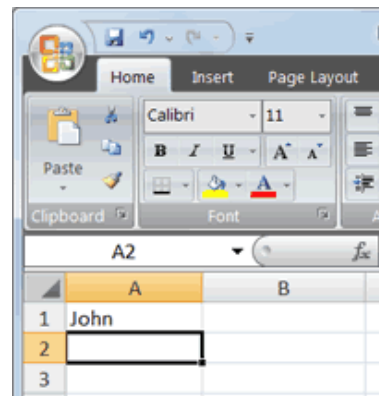
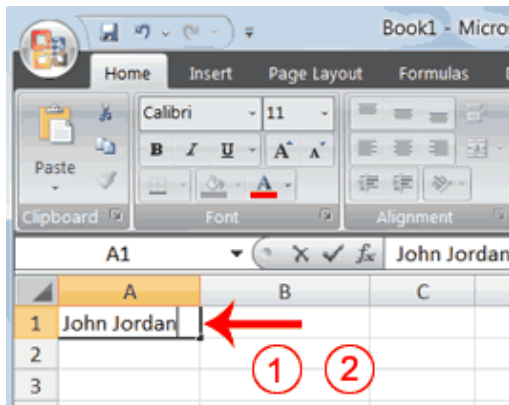
1. Go to cell A1.
2. Hold down the Ctrl key. You won't release it until step 9. Holding down the Ctrl key enables you to select noncontiguous areas of the worksheet.
3. Press the left mouse button.
4. While holding down the left mouse button, use the mouse to move from cell A1 to C5.
5. Continue to hold down the Ctrl key, but release the left mouse button.
6. Using the mouse, place the cursor in cell D7.
7. Press the left mouse button.
8. While holding down the left mouse button, move to cell F10. Release the left mouse button.
9. Release the Ctrl key. Cells A1 to C5 and cells D7 to F10 are selected.
10. Press Esc and click anywhere on the worksheet to remove the highlighting.

Enter Data

In this section, you will learn how to enter data into your worksheet. First, place the cursor in the cell in which you want to start entering data. Type some data, and then press Enter. If you need to delete, press the Backspace key to delete one character at a time.

Enter Data

1. Place the cursor in cell A1.
2. Type **John Jordan**. Do not press Enter at this time.



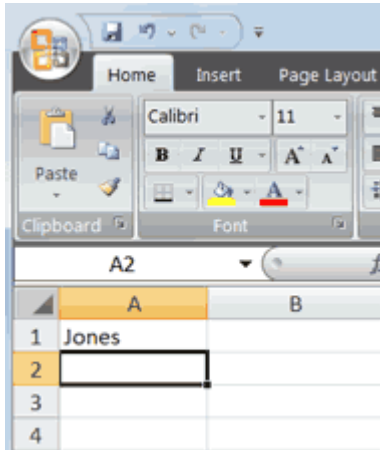
Delete Data

The Backspace key erases one character at a time.

1. Press the Backspace key until Jordan is erased.
2. Press Enter. The name "John" appears in cell A1.

Edit a Cell

After you enter data into a cell, you can edit the data by pressing F2 while you are in the cell you wish to edit.



Edit a Cell

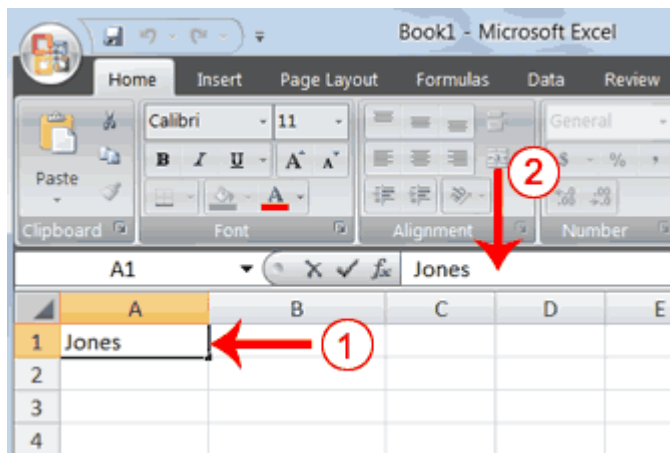
Change "John" to "Jones."

1. Move to cell A1.
2. Press F2.
3. Use the Backspace key to delete the "n" and the "h."
4. Type **nes**.

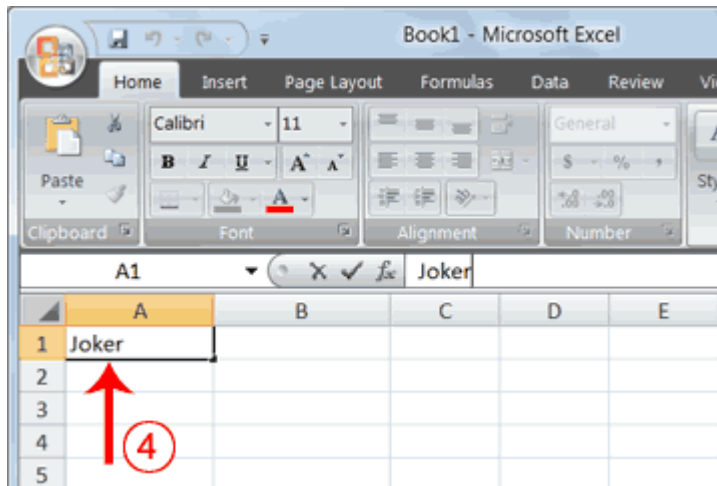
5. Press Enter.

Alternate Method: Editing a Cell by Using the Formula Bar

You can also edit the cell by using the Formula bar. You change "Jones" to "Joker" in the following exercise.



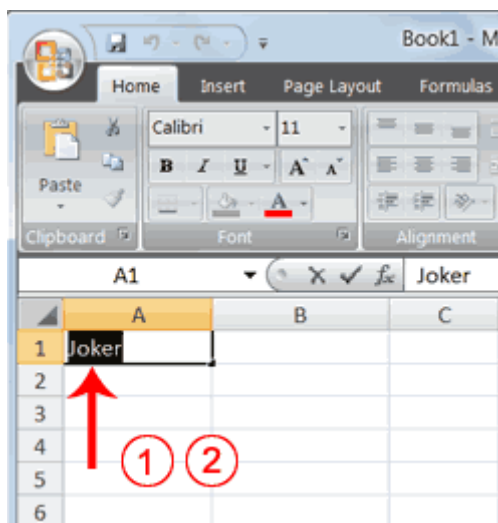
1. Move the cursor to cell A1.
2. Click in the formula area of the Formula bar.



3. Use the backspace key to erase the "s," "e," and "n."
4. Type **ker**.
5. Press Enter.

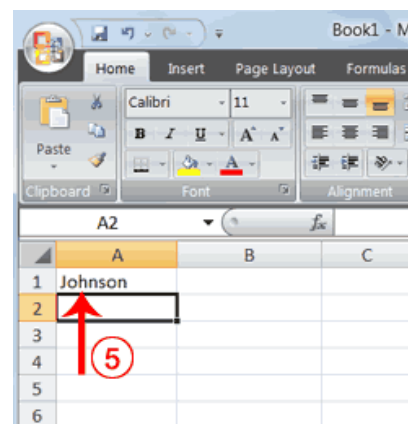
Alternate Method: Edit a Cell by Double-Clicking in the Cell

You can change "Joker" to "Johnson" as follows:



1. Move to cell A1.
2. Double-click in cell A1.

3. Press the End key. Your cursor is now at the end of your text.

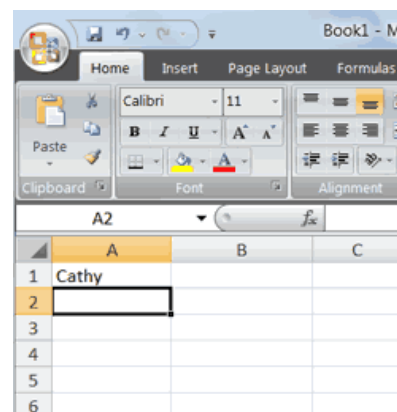


3. Use the Backspace key to erase "r," "e," and "k."
4. Type **hnson**.
5. Press Enter.

Change a Cell Entry

Typing in a cell replaces the old cell entry with the new information you type.

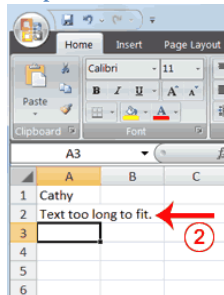
1. Move the cursor to cell A1.
2. Type **Cathy**.
3. Press Enter. The name "Cathy" replaces "Johnson."



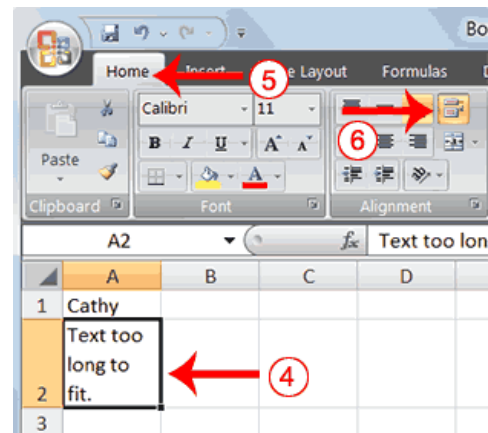
Wrap Text

When you type text that is too long to fit in the cell, the text overlaps the next cell. If you do not want it to overlap the next cell, you can wrap the text.

Wrap Text



1. Move to cell A2.
2. Type **Text too long to fit**.
3. Press Enter.



4. Return to cell A2.
5. Choose the Home tab.
6. Click the Wrap Text button . Excel wraps the text in the cell.

Create Borders

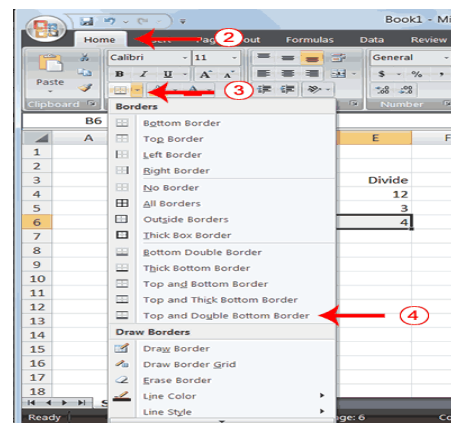
You can use borders to make entries in your Excel worksheet stand out. You can choose from several types of borders.

When you press the down arrow next to the Border button , a menu appears. By making the proper selection from the menu, you can place a border on the top, bottom, left, or right side of the selected cells; on all sides; or around the outside border. You can have a thick outside border or a border with a single-line top and a double-line bottom. Accountants usually place a single underline above a final number and a double underline below. The following illustrates:

Create Borders

	A	B	C	D	E	F	G
1							
2							
3		Add	Subtract	Multiply	Divide		
4		2	8	4	12		
5		1	3	3	3		
6		3	5	12	4		
7							
8							

1. Select cells B6 to E6.



2. Choose the Home tab.

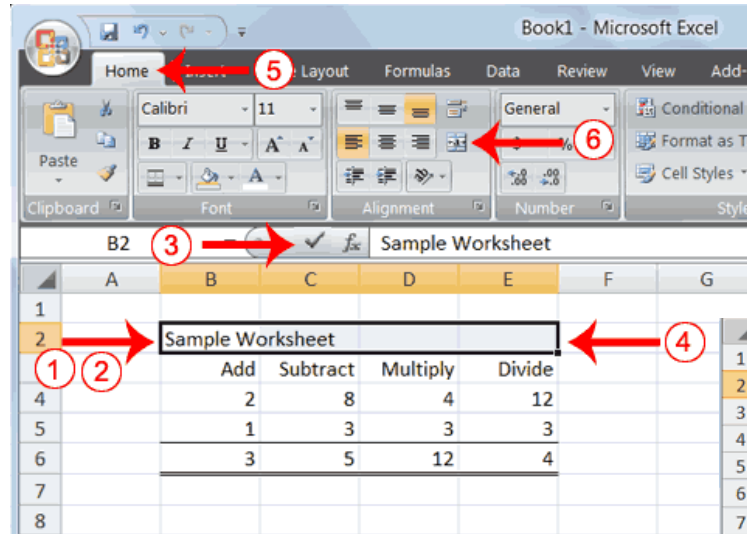
- Click the down arrow next to the Borders button . A menu appears.
- Click Top and Double Bottom Border. Excel adds the border you chose to the selected cells.

	A	B	C	D	E	F	G
1							
2							
3		Add	Subtract	Multiply	Divide		
4		2	8	4	12		
5		1	3	3	3		
6		3	5	12	4		
7							
8							

Merge and Center

Sometimes, particularly when you give a title to a section of your worksheet, you will want to center a piece of text over several columns or rows. The following example shows you how.

Merge and Center



- Type **Sample Worksheet**.
- Click the check mark on the Formula bar.
- Select cells B2 to E2.
- Choose the Home tab.
- Click the Merge and Center button  in the Alignment group. Excel merges cells B2, C2, D2, and E2 and then centers the content.

	A	B	C	D	E	F
1						
2		Sample Worksheet				
3		Add	Subtract	Multiply	Divide	
4		2	8	4	12	
5		1	3	3	3	
6		3	5	12	4	
7						
8						

- Go to cell B2.

Note: To unmerge cells:

- Select the cell you want to unmerge.
- Choose the Home tab.
- Click the down arrow next to the Merge and Center button.  A menu appears.
- Click Unmerge Cells. Excel unmerges the cells.

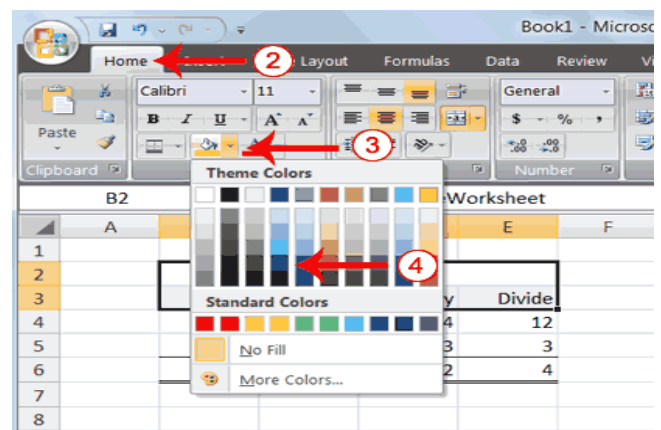
Add Background Color

To make a section of your worksheet stand out, you can add background color to a cell or group of cells.

Add Background Color

	A	B	C	D	E	F
1						
2		Sample Worksheet				
3		Add	Subtract	Multiply	Divide	
4		2	8	4	12	
5		1	3	3	3	
6		3	5	12	4	
7						
8						

- Select cells B2 to E3.
- Choose the Home tab.
- Click the down arrow next to the Fill Color button .
- Click the color dark blue. Excel places a dark blue background in the cells you selected.



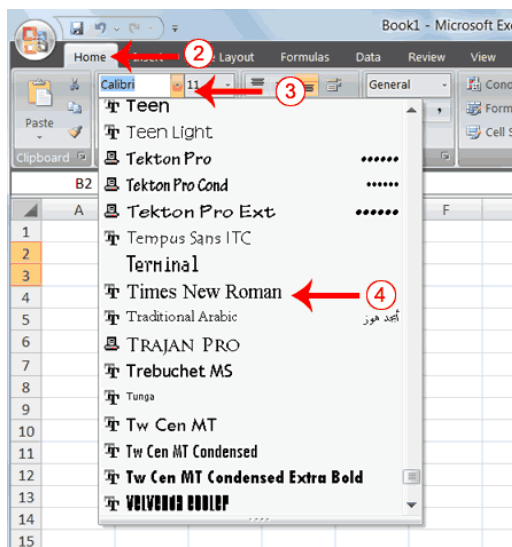
	A	B	C	D	E	F
1						
2		Sample Worksheet				
3		Add	Subtract	Multiply	Divide	
4		2	8	4	12	
5		1	3	3	3	
6		3	5	12	4	
7						
8						

Change the Font, Font Size, and Font Color

A font is a set of characters represented in a single typeface. Each character within a font is created by using the same basic style. Excel provides many different fonts from which you can choose. The size of a font is measured in points. There are 72 points to an inch. The number of points assigned to a font is based on the distance from the top to the bottom of its longest character. You can change the Font, Font Size, and Font Color of the data you enter into Excel.

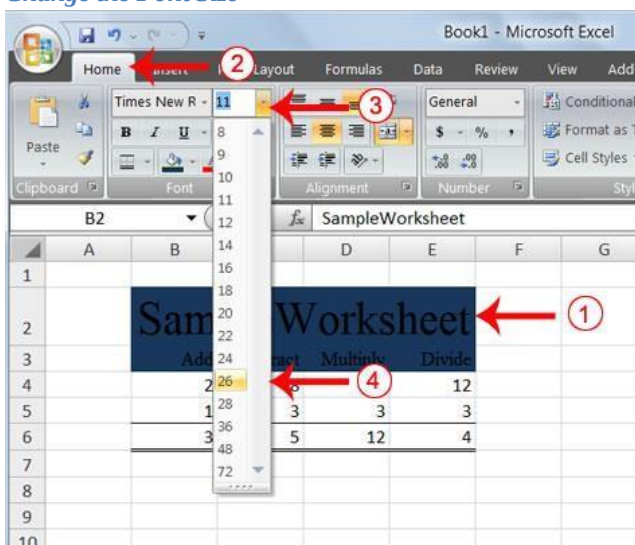
Change the Font

1. Select cells B2 to E3.



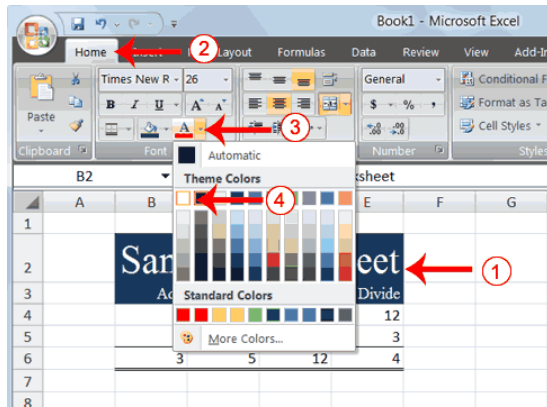
2. Choose the Home tab.
3. Click the down arrow next to the Font box. A list of fonts appears. As you scroll down the list of fonts, Excel provides a preview of the font in the cell you selected.
4. Find and click Times New Roman in the Font box. **Note:** If Times New Roman is your default font, click another font. Excel changes the font in the selected cells.

Change the Font Size



1. Select cell B2.
2. Choose the Home tab.
3. Click the down arrow next to the Font Size box. A list of font sizes appears. As you scroll up or down the list of font sizes, Excel provides a preview of the font size in the cell you selected.
4. Click 26. Excel changes the font size in cell B2 to 26.

Change the Font Color



1. Select cells B2 to E3.
2. Choose the Home tab.

3. Click the down arrow next to the Font Color button .
4. Click on the color white. Your font color changes to white.

	A	B	C	D	E	F
1						
2		Sample Worksheet				
3		Add	Subtract	Multiply	Divide	
4		2	8	4	12	
5		1	3	3	3	
6		3	5	12	4	
7						
8						

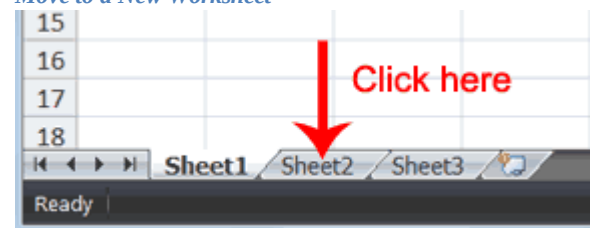
Your worksheet should look like the one shown here

Move to a New Worksheet

In Microsoft Excel, each workbook is made up of several worksheets. Each worksheet has a tab. By default, a workbook has three sheets and they are named sequentially, starting with Sheet1. The name of shows you how.

the worksheet appears on the tab. Before moving to the next topic, move to a new worksheet. The exercise that follows

Move to a New Worksheet

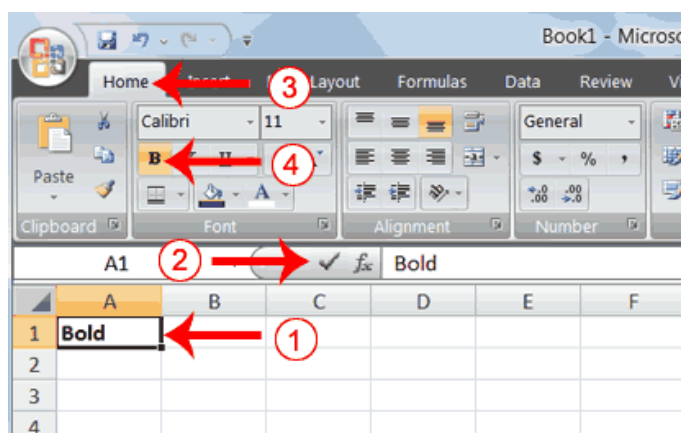


- Click Sheet2 in the lower-left corner of the screen. Excel moves to Sheet2.

Bold, Italicize, and Underline

When creating an Excel worksheet, you may want to emphasize the contents of cells by bolding, italicizing, and/or underlining. You can easily bold, italicize, or underline text with Microsoft Excel. You can also combine these features—in other words, you can bold, italicize, and underline a single piece of text. In the exercises that follow, you will learn different methods you can use to bold, italicize, and underline.

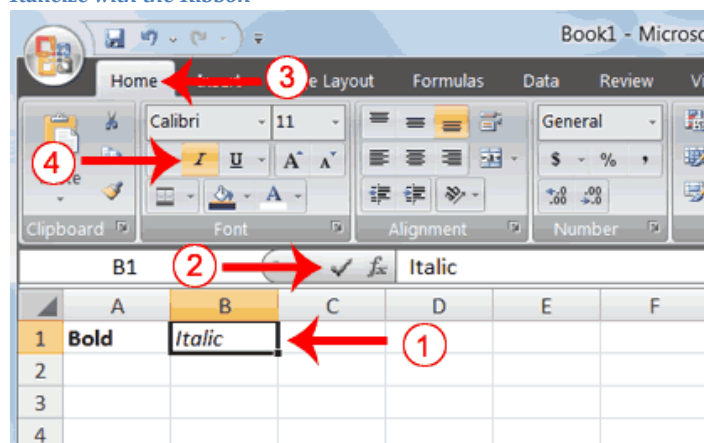
Bold with the Ribbon



1. Type **Bold** in cell A1.

2. Click the check mark located on the Formula bar.
3. Choose the Home tab.
4. Click the Bold button . Excel bolds the contents of the cell.
5. Click the Bold button  again if you wish to remove the bold.

Italicize with the Ribbon

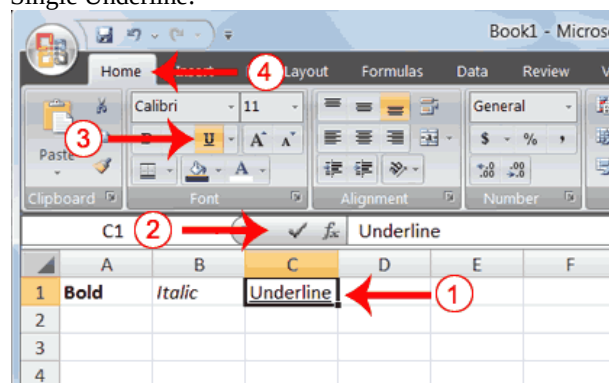


1. Type **Italic** in cell B1.
2. Click the check mark located on the Formula bar.
3. Choose the Home tab.
4. Click the Italic button . Excel italicizes the contents of the cell.
5. Click the Italic button  again if you wish to remove the italic.

Underline with the Ribbon

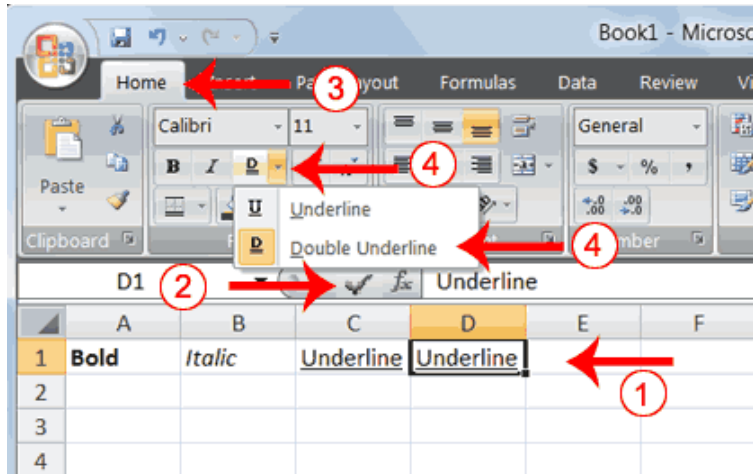
Microsoft Excel provides two types of underlines. The exercises that follow illustrate them.

Single Underline:



1. Type **Underline** in cell C1.
2. Click the check mark located on the Formula bar.
3. Choose the Home tab.
4. Click the Underline button . Excel underlines the contents of the cell.
5. Click the Underline button  again if you wish to remove the underline.

Double Underline



1. Type **Underline** in cell D1.

2. Click the check mark located on the Formula bar.
3. Choose the Home tab.
4. Click the down arrow next to the Underline button **U** and then click Double Underline. Excel double-underlines the contents of the cell. Note that the Underline button changes to the button shown here **D**, a D with a double underline under it. Then next time you click the Underline button, you will get a double underline. If you want a single underline, click the down arrow next to the Double Underline button **D** and then choose Underline.
5. Click the double underline button again if you wish to remove the double underline.

Bold, Underline, and Italicize

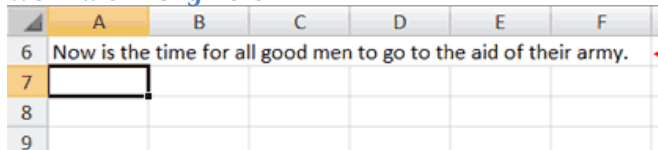
1. Type **All three** in cell E1.
2. Click the check mark located on the Formula bar.
3. Choose the Home tab.
4. Click the Bold button **B**. Excel bolds the cell contents.

5. Click the Italic button **I**. Excel italicizes the cell contents.
6. Click the Underline button **U**. Excel underlines the cell contents.

Work with Long Text

Whenever you type text that is too long to fit into a cell, Microsoft Excel attempts to display all the text. It left-aligns the text regardless of the alignment you have assigned to it, and it borrows space from the blank cells to the right. However, a long text entry will never write over cells that already contain entries—instead, the cells that contain entries cut off the long text. The following exercise illustrates this.

Work with Long Text

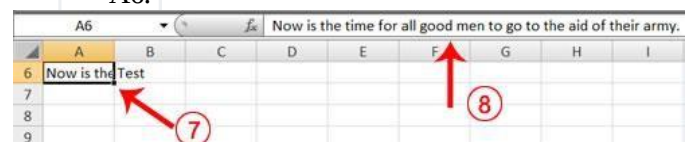


1. Move to cell A6.
2. Type **Now is the time for all good men to go to the aid of their army.**
3. Press Enter. Everything that does not fit into cell A6 spills over into the adjacent cell.

4. Move to cell B6.

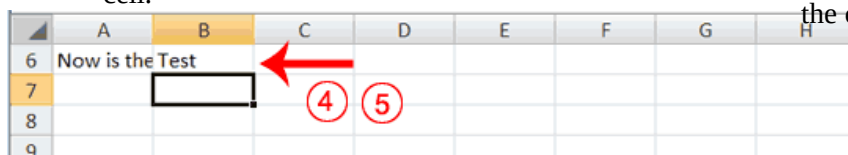
5. Type **Test.**

6. Press Enter. Excel cuts off the entry in cell A6.



7. Move to cell A6.

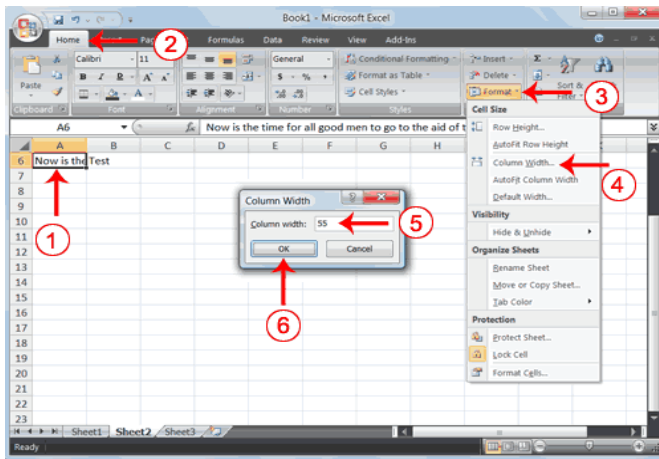
8. Look at the Formula bar. The text is still in the cell.



Change A Column's Width

You can increase column widths. Increasing the column width enables you to see the long text.

Change Column Width

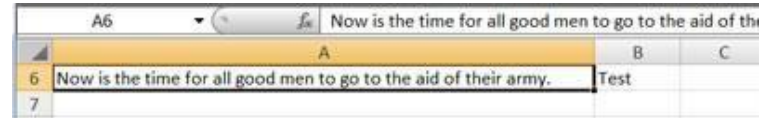


1. Make sure you are in any cell under column A.

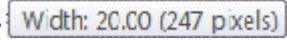
You can also change the column width with the cursor.

1. Place the mouse pointer on the line between the B and C column headings. The mouse pointer should look like the one displayed here , with two arrows.
2. Move your mouse to the right while holding down the left mouse button. The width

2. Choose the Home tab.
3. Click the down arrow next to Format in the Cells group.
4. Click Column Width. The Column Width dialog box appears.
5. Type 55 in the Column Width field.
6. Click OK. Column A is set to a width of 55. You should now be able to see all of the text.



Change a Column Width by Dragging

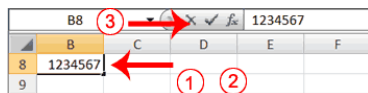
indicator  appears on the screen.

3. Release the left mouse button when the width indicator shows approximately 20. Excel increases the column width to 20.

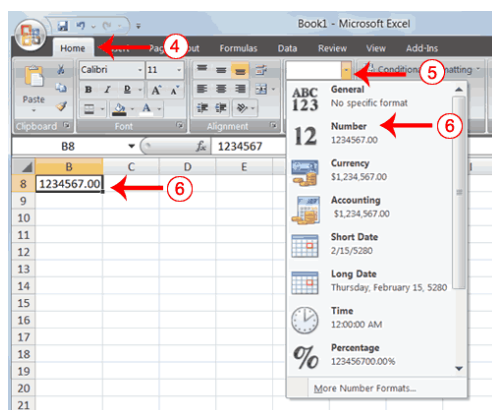
Format Numbers

You can format the numbers you enter into Microsoft Excel. For example, you can add commas to separate thousands, specify the number of decimal places, place a dollar sign in front of a number, or display a number as a percent.

Format Numbers

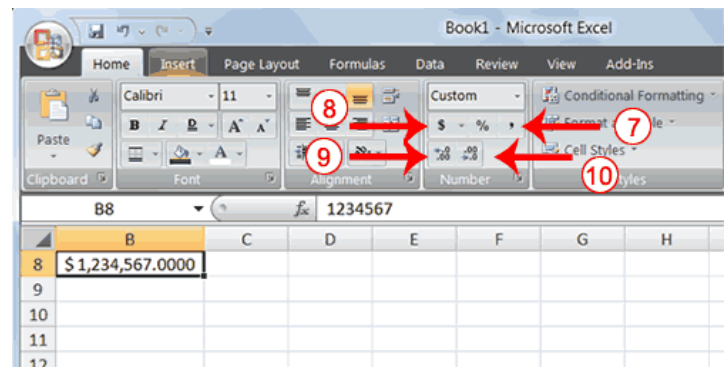


1. Move to cell B8.
2. Type 1234567.
3. Click the check mark on the Formula bar.



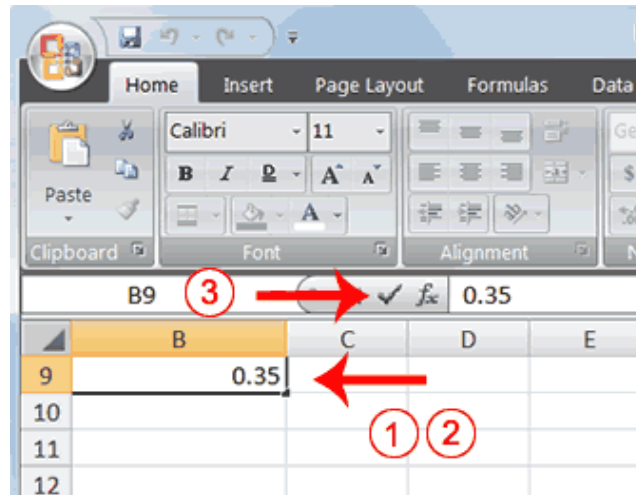
4. Choose the Home tab.
5. Click the down arrow next to the Number Format box. A menu appears.

6. Click Number. Excel adds two decimal places to the number you typed.

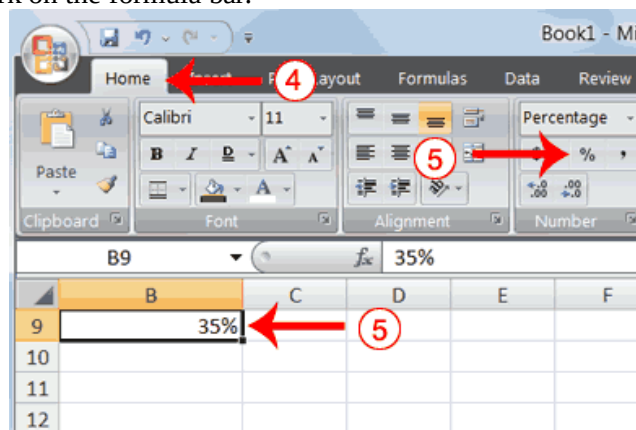


7. Click the Comma Style button . Excel separates thousands with a comma.
8. Click the Accounting Number Format button . Excel adds a dollar sign to your number.
9. Click twice on the Increase Decimal button  to change the number format to four decimal places.
10. Click the Decrease Decimal button  if you wish to decrease the number of decimal places.

Change a decimal to a percent.



1. Move to cell B9.
2. Type .35 (note the decimal point).
3. Click the check mark on the formula bar.



4. Choose the Home tab.
5. Click the Percent Style button . Excel turns the decimal to a percent.

Creating Excel Functions and Filling Cells

By using functions, you can quickly and easily make many useful calculations, such as finding an average, the highest number, the lowest number, and a count of the number of items in a list. Microsoft Excel has many functions that you can use.

Using Reference Operators

To use functions, you need to understand reference operators. Reference operators refer to a cell or a group of cells. There are two types of reference operators: *range* and *union*.

A range reference refers to all the cells between and including the reference. A range reference consists of two cell addresses separated by a colon. The reference A1:A3 includes cells A1, A2, and A3. The reference A1:C3 includes cells A1, A2, A3, B1, B2, B3, C1, C2, and C3.

A union reference includes two or more references. A union reference consists of two or more numbers, range references, or cell addresses separated by a comma. The reference A7,B8:B10,C9,10 refers to cells A7, B8 to B10, C9 and the number 10.

Working with Data

Excel workbooks are designed to store a lot of information. Whether working with 20 cells or 20,000, Excel has several features to help users organize data and find what is needed.

Freezing Rows and Columns

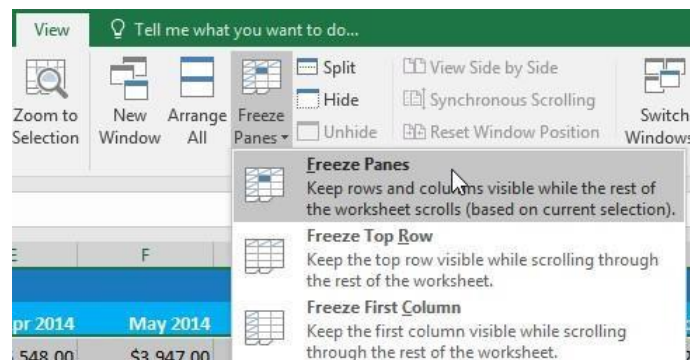
By freezing rows or columns in place, especially header cells, users can scroll through the content while continuing to view the frozen cells. By freezing rows or columns in place, especially header cells, users can scroll through the content while continuing to view the frozen cells.

	A	B	C	D	E	F
1	2014-2015 Sales Data					
2	Salesperson	Jan 2014	Feb 2014	Mar 2014	Apr 2014	May 2014
18	Hodges, Melissa	\$4,624.00	\$14,772.00	\$19,830.00	\$6,303.00	\$5,667.00
19	Jameson, Robinson	\$2,552.00	\$1,627.00	\$4,382.00	\$9,083.00	\$4,269.00
20	Kellerman, France	\$4,281.00	\$7,375.00	\$17,730.00	\$19,998.00	\$3,502.00
21	Mark, Katharine	\$4,679.00	\$3,058.00	\$1,497.00	\$5,722.00	\$5,853.00

To practice with the MS Excel environment

Freezing rows/columns is useful when certain headers (rows/columns) must always be visible as the content is scrolled through.

To freeze rows/columns, select the row/column below/ after the row/column that will be frozen. On the **View** tab, select the **Freeze Panes** command, then choose **Freeze Panes**



from the drop-down menu. Frozen rows/columns will be indicated by a gray line.

To unfreeze, click the **Freeze Panes** command, then select **Unfreeze Panes** from the dropdown menu.

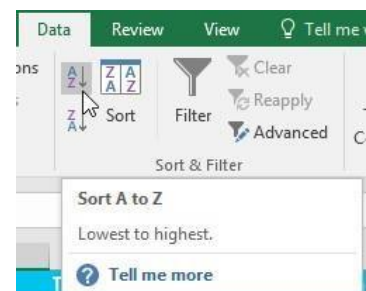
Sorting Data

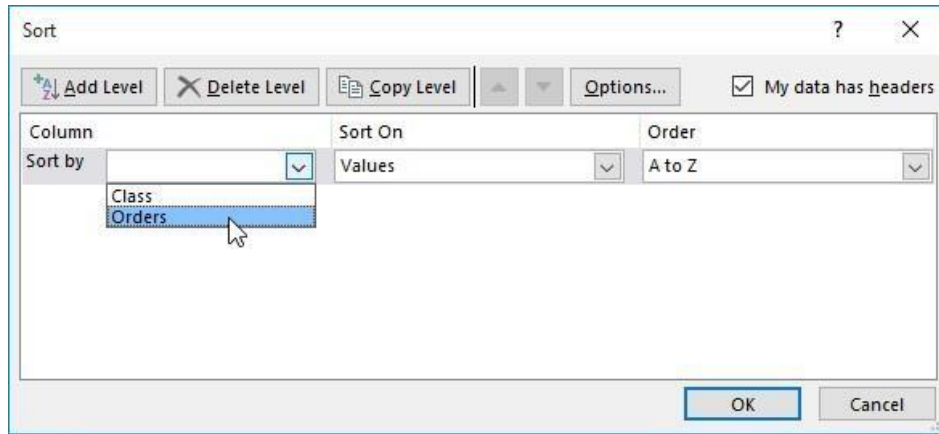
A worksheet can be sorted alphabetically, numerically, and in several other ways. When sorting data, it is important to first decide if sorting should be applied to the entire worksheet or just a cell range.

Sort sheet organizes all the data in the worksheet by one column. Related information across each row is kept together when the sort is applied. Select a cell in the column to sort. Select the **Data** tab on the Ribbon, then click the **A-Z** command to sort A to Z (ascending order), or the **Z-A** command to sort Z to A (descending order). The worksheet will be sorted by the selected column.

Sort range sorts the data in a range of cells, which can be helpful when working with a sheet that contains several tables. Sorting a range will not affect other content in the worksheet. Select the cell range to sort. Select the **Data** tab on the Ribbon, then click the **Sort** command. The *Sort* dialog box will appear.

Choose the column to sort. Decide the sorting order (either ascending or descending). Click **OK**. The cell range will be sorted by the selected column.





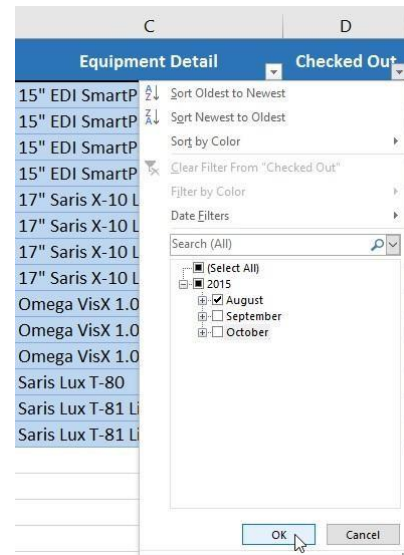
Filtering Data

Filters can be used to narrow down the data in the worksheet, allowing users to view only the information needed. In order for filtering to work correctly, the worksheet should include a header row, which is used to identify the name of each column.

To filter the data, select the **Data** tab, then click the **Filter** command. A drop-down arrow will appear in the header cell for each column. Click the drop-down arrow for the column to filter. In the *Filter* menu, check the boxes next to the data to filter, then click **OK**. The data will be filtered, temporarily hiding any content that doesn't match the criteria.

Filters are cumulative, which means multiple filters can be applied to help narrow down the results.

To clear a filter, click the drop-down arrow for the filter to clear. In the *Filter* menu, choose **Clear Filter From [COLUMN NAME]** from the *Filter* menu. The filter will be cleared from the column. The previously hidden data will be displayed.



Lab Session 06

OBJECT

Working with Formulas conditions, logical & Trigonometric functions in Microsoft Excel
Entering Formulas

Format worksheet

	A	B	C	D
1		12	150	
2		27	85	
3		24	65	
4	Sum	63	300	
5				
6				

1. Move to cell A4.
2. Type the word **Sum**.
3. Select cells B4 to C4.
4. Choose the Home tab.
5. Click the down arrow next to the Borders button .

you can also calculate a sum by using the **AutoSum** button .

Calculate an Average

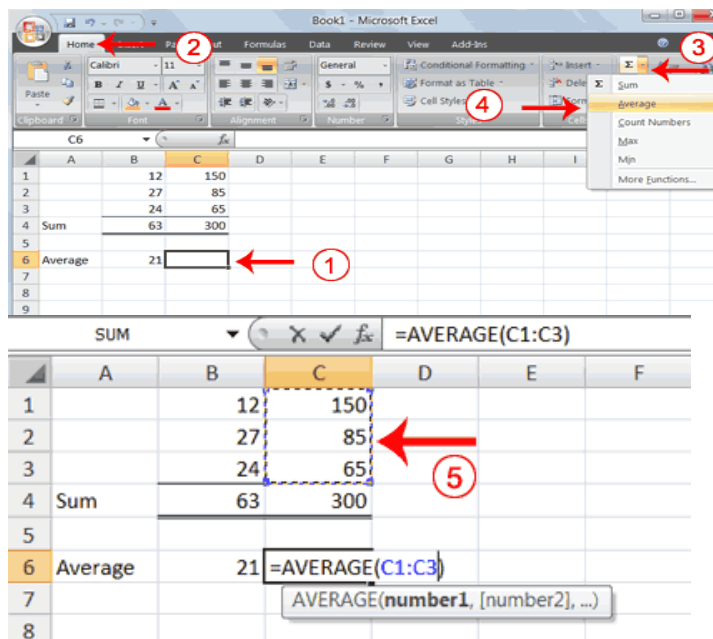
You can use the AVERAGE function to calculate the average of a series of numbers.

B6						=AVERAGE(B1:B3)	
	A	B	C	D	E	F	
1		12	150				
2		27	85				
3		24	65				
4	Sum	63	300				
5							
6	Average	21					
7							

1. Move to cell A6.
2. Type **Average**. Press the right arrow key to move to cell B6.
3. Type **=AVERAGE(B1:B3)**.
4. Press Enter. The average of cells B1 to B3, which is 21, appears.

Calculate an Average with the AutoSum Button

In Microsoft Excel, you can use the AutoSum button  to calculate an average.



The screenshot illustrates the process of using the AutoSum button to calculate an average. The top part shows the Excel ribbon with the Home tab selected, and the AutoSum button (Σ) highlighted. A dropdown menu is open, showing the 'Average' option. The bottom part shows the worksheet with cells C1 to C3 selected, and the formula bar displaying '=AVERAGE(C1:C3)'. Red arrows and numbers 1 through 5 indicate the sequence of steps: 1. Click the AutoSum button. 2. Click the Home tab. 3. Click the down arrow next to the AutoSum button. 4. Click Average. 5. Select cells C1 to C3.

1. Move to cell C6.
2. Choose the Home tab.
3. Click the down arrow next to the AutoSum button .
4. Click Average.
5. Select cells C1 to C3.
6. Press Enter. The average of cells C1 to C3, which is 100, appears.

Find the Minimum Number

You can use the MIN function to find the lowest number in a series of numbers.

	A	B	C	D	E	F
1		12	150			
2		27	85			
3		24	65			
4	Sum	63	300			
5						
6	Average	21	100			
7	Min	12				
8						
9						

1. Move to cell A7.
2. Type **Min**.
3. Press the right arrow key to move to cell B7.
4. Type = **MIN(B1:B3)**.
5. Press Enter. The lowest number in the series, which is 12, appears.

Note: You can also use the drop-down button next to the AutoSum button  to calculate minimums, maximums, and counts.

Find the Maximum Number

You can use the MAX function to find the highest number in a series of numbers.

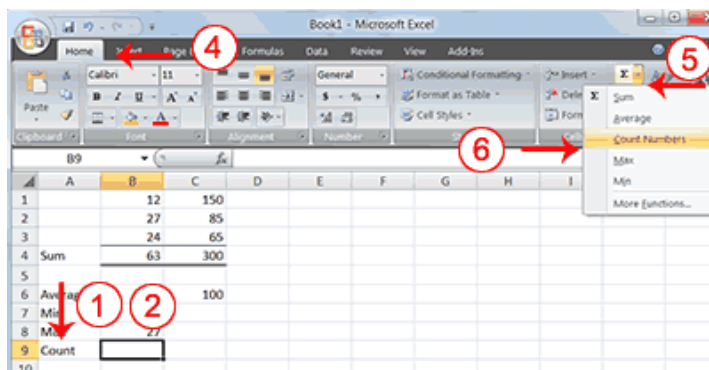
	A	B	C	D	E	F
1		12	150			
2		27	85			
3		24	65			
4	Sum	63	300			
5						
6	Average	21	100			
7	Min	12				
8	Max	27				
9						

1. Move to cell A8.
2. Type **Max**.
3. Press the right arrow key to move to cell B8.
4. Type = **MAX(B1:B3)**.
5. Press Enter. The highest number in the series, which is 27, appears.

6. .

Count the Numbers in a Series of Numbers

You can use the count function to count the number of numbers in a series.



	A	B	C	D	E
1		12	150		
2		27	85		
3		24	65		
4	Sum	63	300		
5					
6	Average	21	100		
7	Min	12			
8	Max	27			
9	Count	=COUNT(B1:B3)			
10		COUNT(value1, [value2], ...)			
11					

7. Select B1 to B3.
8. Press Enter. The number of items in the series, which is 3, appears.

1. Move to cell A9.
2. Type **Count**.
3. Press the right arrow key to move to cell B9.
4. Choose the Home tab.
5. Click the down arrow next to the AutoSum button .
6. Click Count Numbers. Excel places the count function in cell C9 and takes a guess at which cells you want to count. The guess is incorrect, so you must select the proper cells.

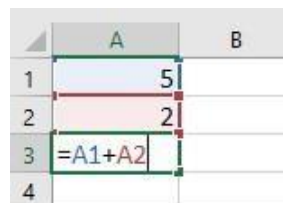
	A	B	C	D	E	F
1		12	150			
2		27	85			
3		24	65			
4	Sum	63	300			
5						
6	Average	21	100			
7	Min	12				
8	Max	27				
9	Count	3				
10						
11						

Formulas and Functions

Formulas

One of the most powerful features in Excel is the ability to calculate numerical information using formulas. Just like a calculator, Excel can add, subtract, multiply, and divide, and uses standard operators for formulas.

- + Addition
- Subtraction
- * Multiplication
- / Division
- ^ Exponents



The image shows a small Excel spreadsheet with two columns, A and B, and four rows, 1 to 4. Cell A1 contains the number 5, and cell A2 contains the number 2. Cell A3 contains the formula =A1+A2. The formula bar at the top shows the formula =A1+A2.

	A	B
1	5	
2	2	
3	=A1+A2	
4		

All formulas in Excel must begin with an equals sign (=). This is because the cell contains, or is equal to, the formula and the value it calculates. Formulas can be created using numbers or cell addresses; the latter is more common. In the example shown, cell A3 will contain the sum of cell A1 and A2. If the content of either A1 and/or A2 is changed, the change will be reflected in cell A3.

Formulas can also include a combination of cell references and numbers, e.g., N10*1.05. Formulas can also be copied to adjacent cells with the fill handle. Formulas can also be edited either in the cell or in the formula bar. Finally, more complex formulas follow the mathematical rules of operator precedence.

Some common Excel formula errors are presented below.

- #####** *Reason:* The column isn't wide enough to display the value.
Solution: Click on the right border of the column header and increase the column width or double-click the right border of the column header to automatically fit the widest entry in that column.
- #NAME?** *Reason:* Excel does not recognize text in a formula. *Solution:* Correct the spelling mistake.
- #VALUE!** *Reason:* A formula has the wrong type of argument.
Solution: Either change the value of cell to a number or use a function to ignore cells that contain text.
- #DIV/0!** *Reason:* A formula tries to divide a number by 0 or an empty cell.
Solution: Either change the value of the cell (in denominator) to a value that is not equal to 0 or prevent the error from being displayed by using the logical function IF.
- #REF!** *Reason:* A formula refers to a cell that is not valid (or deleted afterwards). *Solution:* Either delete +#REF! in the formula of the cell or undo the original deletion.

Functions

A function is a predefined formula that performs calculations using specific values in a particular order. Excel includes many common functions that can be used to quickly find specific results or perform certain calculations.

To work correctly, a function must be written in a specific way.

This is called the syntax. The basic syntax is shown. All functions are composed of function names and a list of arguments. Arguments can refer to both individual cells and cell ranges and must be enclosed within parentheses. Users can include one argument or multiple arguments, depending on the syntax required for the function. Some common functions are presented below.



The diagram shows the basic syntax for a function: =SUM(A1:A20). The equals sign is labeled 'Equals sign'. The word 'SUM' is labeled 'Function name'. The range 'A1:A20' is labeled 'Argument'.

- Average** Returns the Average of a list of supplied numbers.
- AverageIf** Returns the average of cell values (of a supplied range), that satisfies a given criteria.
- Count** Returns the number of numerical values in a supplied set of cells.
- CountIf** Returns the number of cells (of a supplied range), that satisfies a given criteria.
- If** Checks whether a condition is met and returns one value if true and another value if false.
- Max** Returns the largest value from a list of supplied numbers.
- Min** Returns the smallest value from a list of supplied numbers.
- Sum** Returns the sum of a supplied list of numbers.
- SumIf** Returns the sum of cell values (of a supplied range), that satisfies a given criteria.

Functions can also be accessed from the *Function Library* on the **Formulas** tab.

Formulas enable the user to perform calculations by using values in the worksheet. Arithmetic operators that can be used in formulas include + for addition, - for subtraction, * for multiplication, / for division, % for percentage, and ^ for exponentiation. For example: =a1+b1 adds the contents of a1 and b1 in the cell where the formula is typed.

To fill the same formula in multiple cells, select the adjacent cells or ranges to be filled; With the range(s) still selected, type the formula or value in the active cell; Press Ctrl+Enter (rather than just Enter) to enter the formula or value.

To reference cells in other Worksheets, select the cell where the formula is to be appeared , and type an equal sign (=) to start the formula; Click the sheet tab containing the cell to be referenced in the formula; Select the cell or range to be referred to. The complete reference appears in the formula bar; Finish the rest of the formula; then press Enter to complete the formula.

Excel also provides functions like SUM, AVERAGE, etc. Select the cell where the function is to be appeared, and type an equal sign (=) to start the function; Type the function name (such as SUM, AVERAGE, etc) and a left parenthesis; Select the range of cells for the argument and press Enter. Excel automatically adds the closing parenthesis and enters the function. References to columns or rows can also be entered manually, using commas (,) for separation. If colon (:) is used between two references then, it acts as range operator, which produces one reference to all the cells between two references, including the two reference. Following are some of the most commonly used formula.

- **SUM: Adds all the numbers in a range of cells.**
Syntax: =SUM(number1, number2,)
- **AVERAGE: Returns the average (arithmetic mean) of the arguments. Syntax:**
=AVERAGE(number1, number2,)
- **COUNT: Counts the number of cells that contain numbers and numbers within the list of arguments.**
Syntax: =COUNT(value1, value2,)
- **COUNTA: Counts the number of cells that are not empty and the values within the list of arguments.**
Syntax: =COUNTA(value1, value2,)
- **PRODUCT: Multiplies all the numbers given as arguments and returns the product.**
Syntax: =PRODUCT(number1, number2,)

Data Validation

Use data validation in Excel to make sure that users enter certain values into a cell.

Steps:

1. Select a cell.
2. On the Data tab, in the Data Tools group, click Data Validation.
3. On the Settings tab, specify the validation criteria.
4. If you optionally want to show a message (tooltip) on the selected cell, check 'Show input message when cell is selected' and add a message accordingly.
5. If a user ignores the input message and enters a number that is not valid, an error alert can be shown. Check 'Show error alert after invalid data is entered' and add an error message accordingly

SUM PRODUCT FUNCTION: Calculate the product and then return the total sum of the product.

	A	B	C
1	Array 1	Array 2	Multiply
2	3	2	6
3	8	6	48
4	1	5	5
5			59
6	Formula	Description	Result
7	=SUMPRODUCT(A2:A4,B2:B4)	Multiplies all the components of the two	=SUMPRODUCT(A2:A4,B2:B4)

SUMIF: Adds the cells specified by a given criteria.

Syntax: SUMIF(range, criteria, sum_range)

Where 'Range' is the range of cells to be evaluated according to the given criteria. 'Criteria' is the criteria in the form of a number, expression, or text that defines which cells will be added. 'Sum_range' are the actual cells to sum. The cells in sum_range are summed only if their corresponding cells in range match the criteria. If sum_range is omitted, the cells in range are summed.

EmpID	Emp Name	Designation	BASIC	HRA	DA	MA	PF	Gross Salary	Net Salary
101.00	AAA	Manager	12000	2160.00	14400.00	1000.00	9600.00	7960.00	19960.00
102.00	BBB	CEO	15000	2700.00	18000.00	1000.00	12000.00	9700.00	24700.00
103.00	CCC	Accountant	11000	1980.00	13200.00	1000.00	8800.00	7380.00	18380.00
104.00	DDD	Clerk	3000	540.00	3600.00	500.00	2400.00	2240.00	5240.00
105.00	EEEE	Peon	1500	270.00	1800.00	500.00	1200.00	1370.00	2870.00
106.00	FFFF	Manager	13000	2340.00	15600.00	1000.00	10400.00	8540.00	21540.00
107.00	GGGG	Peon	2000	360.00	2400.00	500.00	1600.00	1660.00	3660.00
108.00	HHHH	Clerk	3500	630.00	4200.00	500.00	2800.00	2530.00	6030.00
							Maximum	24700.00	
							Minimum	2870.00	
							Average	12797.50	

Count: 2
Designation wise Total Salary: 2

Formula: =SUMIF(C3:C10,"Manager",J3:J10)
SUMIF(range, criteria, [sum_range])

How to use Sumifs, Countifs

How to use **Sumifs**, **Countifs** and **Averageifs**

CONCATENATE (= X ✓ f_x) =sumifs(D4:D22,A4:A22,G6,B4:B22,H6)

	A	B	C	D	E	F	G	H	I	J	K	L	M
1	SUMIFS, AVERAGEIFS & COUNTIFS												
2													
3	Month	Product	Country	Sales Revenue									
4	Jan	Shirt white	USA	546									
5	Jan	Shirt blue	USA	519									
6	Feb	Shirt white	USA	492									
7	Feb	Shirt blue	USA	559									
8	Feb	Shirt white	USA	591									
9	Feb	Shirt white	USA	535									
10	Feb	Shirt blue	USA	550									
11	Feb	Shirt yellow	USA	517									
12	Feb	Shirt blue	UK	449									
13	Feb	Shirt white	UK	410									
14	Mar	Shirt white	UK	435									
15	Mar	Shirt yellow	USA	468									
16	Apr	Shirt blue	USA	568									

Find total sales for the below criteria:

Feb			4103
Feb	Shirt white		=sumifs(D4:D22,A4:A22,G6,B4:B22,H6)
Feb	Shirt white	USA	

Find out how many shirts were sold for the below criteria:

Feb			
Feb	Shirt white		
Feb	Shirt white	USA	

Formulas: Absolute, Relative, and Mixed Cell References

There are two types of cell references: Relative and Absolute. Both references behave differently when copied and filled to other cells.

Relative references

Relative references change when a formula is copied across multiple cells. By default, all cell references are relative references i.e. they change based on the relative position of rows and columns. For example, if you copy the formula =A2+B2 from row 1 to row 2, the formula will become =A3+B3 automatically. Relative references are especially convenient whenever you need to repeat the same calculation across multiple rows or columns.

Absolute references

There may be times when you do not want a cell reference to change when filling cells. Unlike relative references, absolute references do not change when copied or filled. You can use an absolute reference to keep a row and/or column constant. An absolute reference is designated in a formula by the addition of a dollar sign (\$). It can precede the column reference, the row reference, or both.

Reference	Description
\$A\$2	The column and the row do not change when copied.
A\$2	The row does not change when copied.
\$A2	The column does not change when copied.

Logical Functions

The logical functions enable the user to add decision-making and logical tests to the worksheets. The IF statement is useful for testing conditions and making decisions based on a cell's contents. The AND and OR functions can test multiple criteria or test conditions for use in IF functions. The following examples show the use:

- 1. =AND(D15,G23<30) result is TRUE only when D15 is not zero and G23 is less than 30
OR =AND C4>B4,D4="YES" RESULT IS TRUE comparison of two values

	A	B	C	D	E
1					
2					
3	Division	Budget	Amt Spent	Growth Potential	Needs Review
4	Engines	\$250,000	\$286,000.00	Yes	
5	Wheels	\$75,000	\$71,245.00	Yes	
6	Axles	\$125,000	\$137,456.00	No	
7	Chassis	\$205,000	\$190,000.00	No	
8					
9					

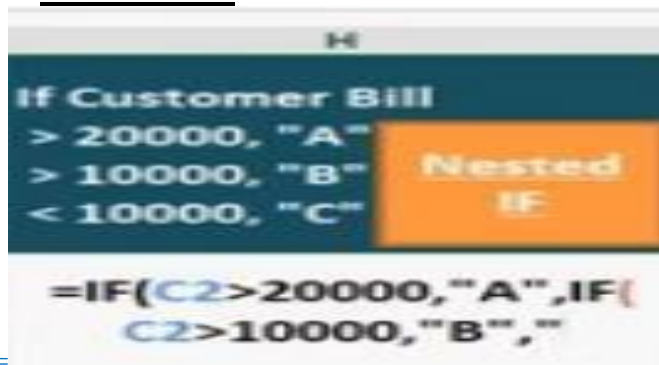
IF:AND

=IF(AND(<Test 1>, <Test 2>, ...), TRUE, FALSE)

- =IF(AND(D17>10,D17<30),"Valid","Invalid") returns Valid if the contents of cell D17 is greater than 10 and less than 30; otherwise the formula returns Invalid

	A	B	C	D	E
1					
2					
3	Needs Review				
4				=IF(AND(C4>B4,D4="no"),"Review ASAP","No Review Needed")	
5					

3. NESTED IF



- = IF(OR(D17=10,D17=30),"Contains 10 or 30","Not 10or30") tests whether cell D17 contains the result 10 or 30 and produces the message Contains 10 or 30 when it does; otherwise, the formula produces the message Not 10 or 30

Math and Trigonometric Functions

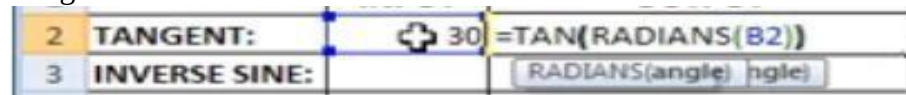
Like logical functions, math and trigonometric functions can also be used in the Worksheets.

Trigonometric functions use angles measured in radians.

Following are some examples:

=Radians(NUMBER) : Convert Degree In To Radian

Tangent:



Sin Function: =SIN(NUMBER) : sin(radians(30))

- **number** - The angle in radians for which you want the sine.

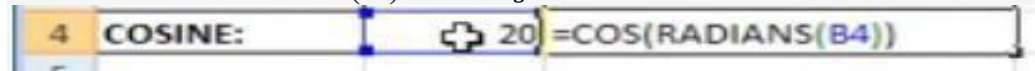
Inverse sine: Result Will Be In Degrees according to formula given below.



Syntax COS

=COS (number) : =Cos(radians(30))

Or =Radians(30) Convert Degree In To Radian



Arguments

- **number** - The angle in radians for which you want the cosine.

ACOS: The Excel ACOS function calculates the arccosine (i.e. the inverse **cosine**) of a given number, and returns an angle, in radians, between 0 and π .

Result In Radian

For degrees =degree(ACOS(RADIAN(DEGREES(30))))

- ☐ =ABS(A10) returns 18 when cell A10 contains -18
- ☐ =ACOS(0.5) returns 1.047198 (radians)
- ☐ =DEGREES(0.5) returns 28.64789 (degrees)
- ☐ =ROUND(102.927,2) returns 102.93
- Mround

Number	Formulae	Description	Result
5.1		Rounds Number to the nearest multiple of 1.	=MROUND(A2,1)
7.5		Rounds Number to the nearest multiple of 5.	MROUND(number, multiple)
24		Rounds Number to the nearest multiple of 10.	

Vlookup and Hlookup Functions

The Vlookup (Vertical lookup) function looks for a value in the leftmost column of a table, and then returns a value in the same row from another column you specify.

Steps:

1. Insert the Vlookup function in another table. A function argument dialog box appears.
2. Enter a value for the Lookup_value field. This field contains the value that is to be found in the first column of the main table, and can be a value, a reference, or a text string.
3. Next, select a set of values for table_array field that contains data to be retrieved. It can be a reference or a range.
4. Then, provide a Col_index_num value. It is the column number in table_array from which the matching value should be returned.
5. Lastly, in the range_lookup field, provide a Boolean value. The boolean False tells the Vlookup function to return an exact match. If the VLOOKUP function cannot find the value in the first column, it will return a #N/A error. The boolean TRUE tells the VLOOKUP function to return an approximate match. If the VLOOKUP function cannot find the value in the first column, it will return the largest value smaller than that number.

If the leftmost column of the table contains duplicates, the Vlookup function matches the first instance only. Also, it performs case-insensitive lookups.

In a similar way, you can use the Hlookup (Horizontal lookup) function.

VLOOKUP and HLOOKUP are functions in Excel that allow users to search a table of data based on what the user has supplied and give appropriate information from that table. *VLOOKUP* allows searching in a table that is set up vertically. All of the data should be set up in columns and each column is responsible for one kind of data. *HLOOKUP* is the exact same function but looks up data that has been formatted by rows instead of columns.

VLOOKUP/HLOOKUP have three arguments in order.

lookup_value	User input. This is the value that the function uses to search on.
table_array	The area of cells in which the table is located. This includes the entire range/table from which the data will be extracted. All columns of the data must be included.
col_index_num	The column of data that contains the answer. For <i>HLOOKUP</i> , this will be <i>row_index_num</i> .
range_lookup	A TRUE or FALSE value. When set to TRUE, the lookup function gives the closest match to the <i>lookup_value</i> without going over the <i>lookup_value</i> . When set to FALSE, an exact match must be found to the <i>lookup_value</i> or the function will return #N/A. Note, this requires that the column containing the <i>lookup_value</i> be formatted in ascending order.

Formula Errors

Some of the common Excel formula errors are as follows:

Error	Description	Solution
##### Error	When your cell contains this error code, the column isn't wide enough to display the value.	Click on the right border of the column header and increase the column width or Double click the right border of the column header to automatically fit the widest entry in that column.
#NAME? error	It occurs when Excel does not recognize text in a formula.	Correct the spelling mistake.
#VALUE! Error	Excel displays this error when a formula has the wrong type of argument.	Either change the value of cell to a number or Use a function to ignore cells that contain text.
#DIV/0! Error	Excel displays this error when a formula tries to divide a number by 0 or an empty cell.	Either change the value of the cell (in denominator) to a value that is not equal to 0 or Prevent the error from being displayed by using the logical function IF.
#REF! error	Excel displays this error when a formula refers to a cell that is not valid (or deleted afterwards).	To fix this error, you can either delete +#REF! in the formula of the cell or you can undo your deletion.

Lab Session 06

OBJECT

Working with Formulas conditions, logical & Trigonometric functions in Microsoft Excel

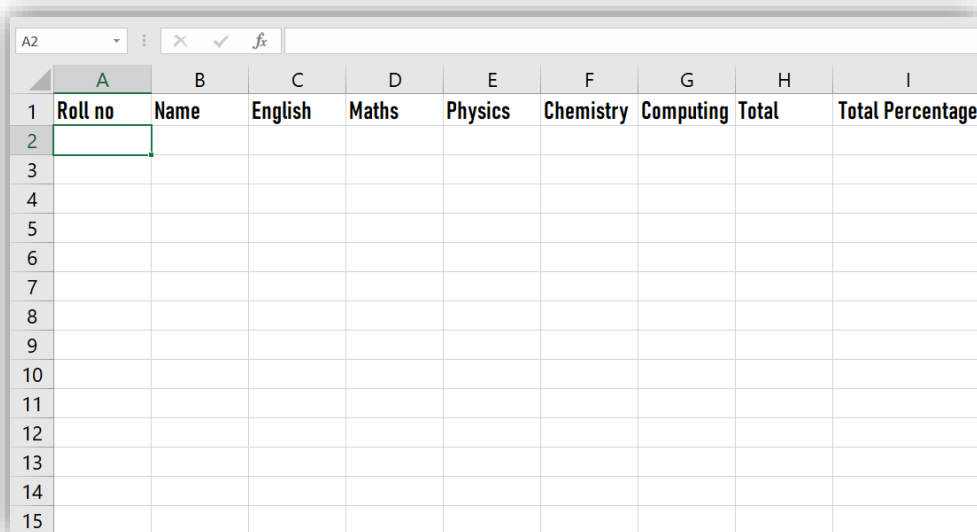
Entering Formulas

EXERCISES

Task 01

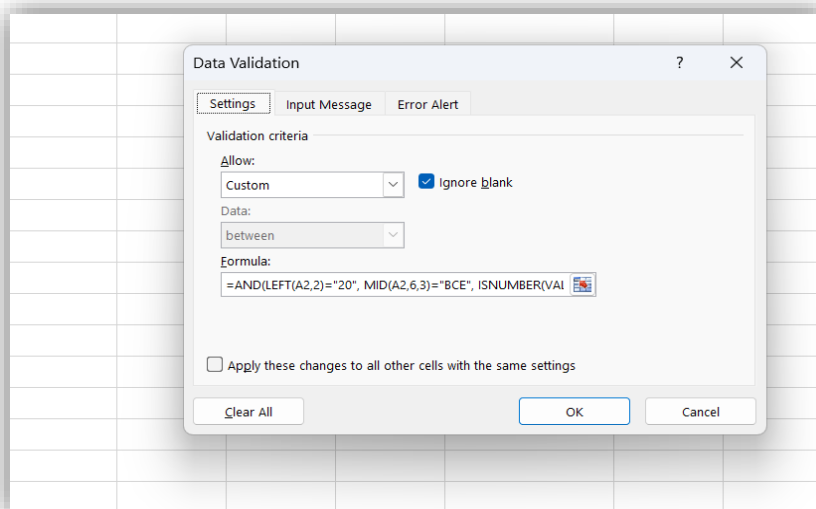
Perform the following operations then print and attach your result.

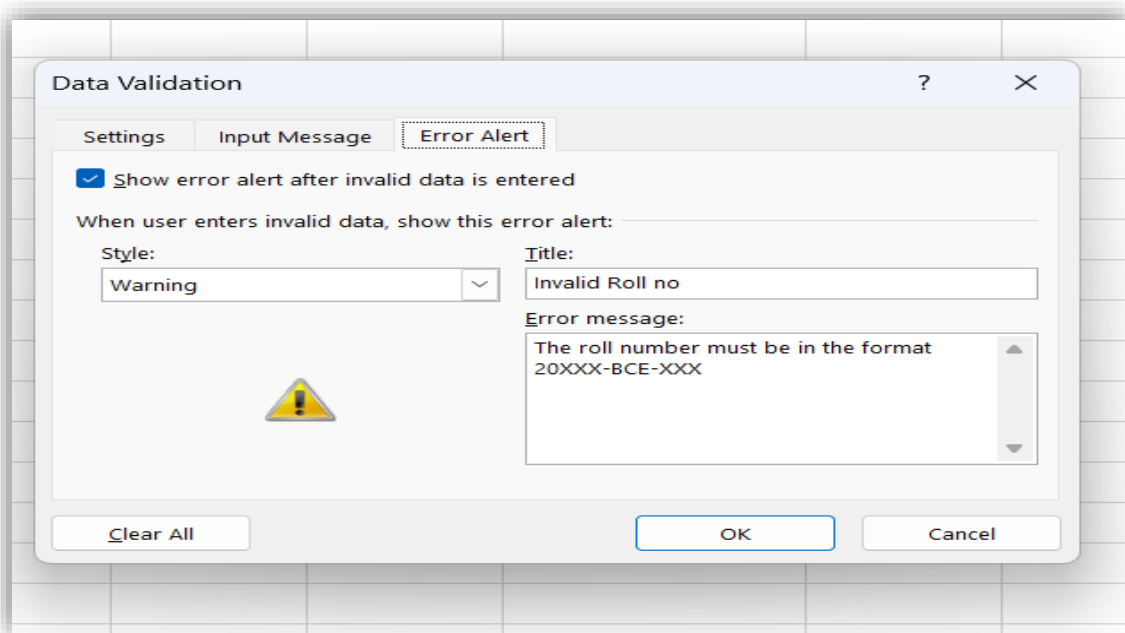
1. Make the following columns in your worksheet: Roll No, Name, English, Math, Physics, Chemistry, Computing, Total, Percentage. Resize the columns if required.



	A	B	C	D	E	F	G	H	I
1	Roll no	Name	English	Maths	Physics	Chemistry	Computing	Total	Total Percentage
2									
3									
4									
5									
6									
7									
8									
9									
10									
11									
12									
13									
14									
15									

2. Apply data validation to ensure that the roll number is of the format 20XXX-BCE-XXX.





3. Enter random roll numbers and student names (at least 10).

	A	B
1	Roll no	Name
2	20201-BCE-024	SARIM
3	20201-BCE-022	WAHAB
4	20201-BCE-029	SOHAIL
5	20201-BCE028	AYESHA
6	20201-BCE-025	FATIMA
7	20201-BCE-027	ZARA
8	20201-BCE026	SARA
9	20201-BCE-030	ALI
10	20201-BCE-021	AHMED
11	20201-BCE023	AYAN

4. Sort the students based on their roll numbers.

	A	B	C	D	E	F	G	H	I	J	K
1	Roll no	Name	English	Maths	Physics	Chemistry	Computing	Total	Total Percentage		
2	20201-BCE-024	SARIM									
3	20201-BCE-022	WAHAB									
4	20201-BCE-029	SOHAIL									
5	20201-BCE028	AYESHA									
6	20201-BCE-025	FATIMA									
7	20201-BCE-027	ZARA									
8	20201-BCE026	SARA									
9	20201-BCE-030	ALI									
10	20201-BCE-021	AHMED									
11	20201-BCE023	AYAN									
12											
13											
14											
15											
16											

Sort

Add Level
Delete Level
Copy Level
Options...
☐ My data has headers

Column
Sort by
Column A
Sort On
Values
Order
A to Z

OK
Cancel

	A	B
1	Roll no	Name
2	20201-BCE-021	AHMED
3	20201-BCE-022	WAHAB
4	20201-BCE023	AYAN
5	20201-BCE-024	SARIM
6	20201-BCE-025	FATIMA
7	20201-BCE026	SARA
8	20201-BCE-027	ZARA
9	20201-BCE028	AYESHA
10	20201-BCE-029	SOHAIL
11	20201-BCE-030	ALI
12		

5. Use the *RANDBETWEEN* function to enter the marks of each subject (0-100).

	A	B	C	D	E	F	G
1	Roll no	Name	English	Maths	Physics	Chemistry	Computing
2	20201-BCE-021	AHMED	=RANDBETWEEN(0, 100)				
3	20201-BCE-022	WAHAB					
4	20201-BCE023	AYAN					
5	20201-BCE-024	SARIM					
6	20201-BCE-025	FATIMA					
7	20201-BCE026	SARA					
8	20201-BCE-027	ZARA					
9	20201-BCE028	AYESHA					
10	20201-BCE-029	SOHAIL					
11	20201-BCE-030	ALI					

	A	B	C	D	E	F	G	H	I
1	Roll no	Name	English	Maths	Physics	Chemistry	Computing	Total	Total Percentage
2	20201-BCE-021	AHMED	44	44	75	71	66		
3	20201-BCE-022	WAHAB	94	6	48	41	4		
4	20201-BCE023	AYAN	35	64	51	94	18		
5	20201-BCE-024	SARIM	70	49	100	53	34		
6	20201-BCE-025	FATIMA	3	21	23	79	31		
7	20201-BCE026	SARA	51	92	71	74	12		
8	20201-BCE-027	ZARA	14	45	61	8	32		
9	20201-BCE028	AYESHA	52	97	12	16	22		
10	20201-BCE-029	SOHAIL	90	39	79	46	1		
11	20201-BCE-030	ALI	82	89	58	69	31		

6. Calculate the total marks of a student using the *SUM* function.

H2									
	A	B	C	D	E	F	G	H	I
1	Roll no	Name	English	Maths	Physics	Chemistry	Computing	Total	Total Percentage
2	20201-BCE-021	AHMED	86	33	36	17	57	229	
3	20201-BCE-022	WAHAB	37	65	60	89	46	297	
4	20201-BCE023	AYAN	31	25	66	32	86	240	
5	20201-BCE-024	SARIM	61	12	100	37	84	294	
6	20201-BCE-025	FATIMA	9	73	27	76	56	241	
7	20201-BCE026	SARA	69	3	78	64	35	249	
8	20201-BCE-027	ZARA	61	37	69	82	38	287	
9	20201-BCE028	AYESHA	40	20	53	27	76	216	
10	20201-BCE-029	SOHAIL	39	1	52	77	96	265	
11	20201-BCE-030	ALI	3	37	58	27	34	159	

7. Calculate the percentage of each student by entering the appropriate formula. Round it to 1 decimal place.

	A	B	C	D	E	F	G	H	I
1	Roll no	Name	English	Maths	Physics	Chemistry	Computing	Total	Total Percentage
2	20201-BCE-021	AHMED	65	70	50	15	59	259	51.8
3	20201-BCE-022	WAHAB	97	100	92	37	1	327	
4	20201-BCE023	AYAN	10	89	96	9	29	233	
5	20201-BCE-024	SARIM	27	75	88	75	95	360	
6	20201-BCE-025	FATIMA	29	53	84	93	53	312	
7	20201-BCE026	SARA	1	14	32	13	81	141	
8	20201-BCE-027	ZARA	74	74	91	71	28	338	
9	20201-BCE028	AYESHA	29	95	60	98	79	361	
10	20201-BCE-029	SOHAIL	34	100	94	65	53	346	
11	20201-BCE-030	ALI	73	51	44	83	49	300	

	A	B	C	D	E	F	G	H	I
1	Roll no	Name	English	Maths	Physics	Chemistry	Computing	Total	Total Percentage
2	20201-BCE-021	AHMED	54	18	84	33	85	274	54.8
3	20201-BCE-022	WAHAB	65	45	84	82	77	353	70.6
4	20201-BCE023	AYAN	59	32	3	64	31	189	37.8
5	20201-BCE-024	SARIM	96	43	41	25	70	275	55
6	20201-BCE-025	FATIMA	23	18	1	71	74	187	37.4
7	20201-BCE026	SARA	73	82	80	3	16	254	50.8
8	20201-BCE-027	ZARA	59	21	53	76	58	267	53.4
9	20201-BCE028	AYESHA	85	84	3	72	56	300	60
10	20201-BCE-029	SOHAIL	63	7	60	5	24	159	31.8
11	20201-BCE-030	ALI	29	51	74	98	48	300	60

8. Find the maximum and minimum marks of English. Then use the fill handle to find the maximum and minimum marks of all the subjects.

	A	B	C	D	E	F	G	H	I
1	Roll no	Name	English	Maths	Physics	Chemistry	Computing	Total	Total Percentage
2	20201-BCE-021	AHMED	15	35	89	35	33	207	41.4
3	20201-BCE-022	WAHAB	15	44	44	56	87	246	49.2
4	20201-BCE023	AYAN	9	92	13	31	28	173	34.6
5	20201-BCE-024	SARIM	39	60	45	78	50	272	54.4
6	20201-BCE-025	FATIMA	20	64	93	0	98	275	55
7	20201-BCE026	SARA	16	18	81	66	22	203	40.6
8	20201-BCE-027	ZARA	84	40	58	36	58	276	55.2
9	20201-BCE028	AYESHA	78	61	32	87	99	357	71.4
10	20201-BCE-029	SOHAIL	54	20	35	8	29	146	29.2
11	20201-BCE-030	ALI	53	96	86	58	2	295	59
12	Max. marks of subject		84	96	93	87	99		
13	Min. marks of subjects		9	18	13	0	2		

9. Use a formula to automatically assign grades based on the following criteria: A: 90-100%, B: 80-90%, C: 70-80%, F:< 70%. (Use logical IF).

	A	B	C	D	E	F	G	H	I	J
1	Roll no	Name	English	Maths	Physics	Chemistry	Computing	Total	Total Percentage	Grade
2	20201-BCE-021	AHMED	55	44	66	74	76	315	63	F
3	20201-BCE-022	WAHAB	67	37	90	59	17	270	54	F
4	20201-BCE023	AYAN	31	52	39	57	94	273	54.6	F
5	20201-BCE-024	SARIM	20	31	52	89	29	221	44.2	F
6	20201-BCE-025	FATIMA	31	100	12	19	75	237	47.4	F
7	20201-BCE026	SARA	14	27	95	91	71	298	59.6	F
8	20201-BCE-027	ZARA	79	75	84	70	45	353	70.6	F
9	20201-BCE028	AYESHA	94	28	47	81	25	275	55	C
10	20201-BCE-029	SOHAIL	15	94	49	46	60	264	52.8	F
11	20201-BCE-030	ALI	88	43	55	60	63	309	61.8	F
12	Max. marks of subject		94	100	95	91	94			
13	Min. marks of subjects		14	27	12	19	17			

10. Apply the following conditional formatting to the data: A: Green, B: Blue, C: Yellow, D: Red.

	A	B	C	D	E	F	G	H	I	J	K
1	Roll no	Name	English	Maths	Physics	Chemistry	Computing	Total	Total Percentage	Grade	
2	20201-BCE-021	AHMED	55	44	66	74	76	315	63	F	
3	20201-BCE-022	WAHAB	67	37	90	59	17	270	54	F	
4	20201-BCE023	AYAN	31	52	39	57	94	273	54.6	F	
5	20201-BCE-024	SARIM	20	31	52	89	29	221	44.2	F	
6	20201-BCE-025	FATIMA	31	100	12	19	75	237	47.4	F	
7	20201-BCE026	SARA	14	27	95	91	71	298	59.6	F	
8	20201-BCE-027	ZARA	79	75	84	70	45	353	70.6	F	
9	20201-BCE028	AYESHA	94	28	47	81	25	275	55	C	
10	20201-BCE-029	SOHAIL	15	94	49	46	60	264	52.8	F	
11	20201-BCE-030	ALI	88	43	55	60	63	309	61.8	F	
12	Max. marks of subject		94	100	95	91	94				
13	Min. marks of subjects		14	27	12	19	17				

	A	B	C	D	E	F	G	H	I	J	K
1	Roll no	Name	English	Maths	Physics	Chemistry	Computing	Total	Total Percentage	Grade	
2	20201-BCE-021	AHMED	55	44	66	74	76	315	63	F	
3	20201-BCE-022	WAHAB	67	37	90	59	17	270	54	F	
4	20201-BCE023	AYAN	31	52	39	57	94	273	54.6	F	
5	20201-BCE-024	SARIM	20	31	52	89	29	221	44.2	F	
6	20201-BCE-025	FATIMA	31	100	12	19	75	237	47.4	F	
7	20201-BCE026	SARA	14	27	95	91	71	298	59.6	F	
8	20201-BCE-027	ZARA	79	75	84	70	45	353	70.6	F	
9	20201-BCE028	AYESHA	94	28	47	81	25	275	55	C	
10	20201-BCE-029	SOHAIL	15	94	49	46	60	264	52.8	F	
11	20201-BCE-030	ALI	88	43	55	60	63	309	61.8	F	
12	Max. marks of subject		94	100	95	91	94				
13	Min. marks of subjects		14	27	12	19	17				

Format Cells dialog box (Fill tab) is open over a spreadsheet. The spreadsheet data is as follows:

	A	B	C	D
1	Roll no	Name	English	Maths
2	20201-BCE-021	AHMED	55	44
3	20201-BCE-022	WAHAB	67	37
4	20201-BCE023	AYAN	31	52
5	20201-BCE-024	SARIM	20	31
6	20201-BCE-025	FATIMA	31	100
7	20201-BCE026	SARA	14	27
8	20201-BCE-027	ZARA	79	75
9	20201-BCE028	AYESHA	94	28
10	20201-BCE-029	SOHAIL	15	94
11	20201-BCE-030	ALI	88	43
12	Max. marks of subject		94	100
13	Min. marks of subjects		14	27

The Format Cells dialog box (Fill tab) shows the Background Color set to a bright green color. The Pattern Color is set to Automatic. The Sample area shows a green bar.

New Formatting Rule dialog box is open over a spreadsheet. The spreadsheet data is as follows:

	A	B	C	D	E	F	G	H	I	J
1	Roll no	Name	English	Maths						
2	20201-BCE-021	AHMED	55	44						
3	20201-BCE-022	WAHAB	67	37						
4	20201-BCE023	AYAN	31	52						
5	20201-BCE-024	SARIM	20	31						
6	20201-BCE-025	FATIMA	31	100						
7	20201-BCE026	SARA	14	27						
8	20201-BCE-027	ZARA	79	75						
9	20201-BCE028	AYESHA	94	28						
10	20201-BCE-029	SOHAIL	15	94						
11	20201-BCE-030	ALI	88	43						
12	Max. marks of subject		94	100	95	91	94			
13	Min. marks of subjects		14	27	12	19	17			

The New Formatting Rule dialog box shows the Rule Type set to "Use a formula to determine which cells to format". The Rule Description is "Format only cells with: Specific Text containing 8". The Preview area shows "No Format Set".

Format Cells dialog box (Fill tab) is open over a spreadsheet. The spreadsheet data is as follows:

	A	B	C	D
1	Roll no	Name	English	Maths
2	20201-BCE-021	AHMED	55	
3	20201-BCE-022	WAHAB	67	
4	20201-BCE023	AYAN	31	
5	20201-BCE-024	SARIM	20	
6	20201-BCE-025	FATIMA	31	1
7	20201-BCE026	SARA	14	
8	20201-BCE-027	ZARA	79	
9	20201-BCE028	AYESHA	94	
10	20201-BCE-029	SOHAIL	15	
11	20201-BCE-030	ALI	88	
12	Max. marks of subject		94	1
13	Min. marks of subjects		14	

The Format Cells dialog box (Fill tab) shows the Background Color set to a bright blue color. The Pattern Color is set to Automatic. The Sample area shows a blue bar.

Clipboard Font Alignment Number Styles Cells

2

Clipboard Font Alignment Number Styles Cells

2

1 Roll no Name English Maths

2 20201-BCE-021 AHMED 55 44

3 20201-BCE-022 WAHAB 67 37

4 20201-BCE023 AYAN 31 52

5 20201-BCE-024 SARIM 20 31

6 20201-BCE-025 FATIMA 31 100

7 20201-BCE026 SARA 14 27

8 20201-BCE-027 ZARA 79 75

9 20201-BCE028 AYESHA 94 28

10 20201-BCE-029 SOHAIL 15 94

11 20201-BCE-030 ALI 88 43

12 Max. marks of subject 94 100 95 91 94

13 Min. marks of subjects 14 27 12 19 17

14

15

New Formatting Rule

Select a Rule Type:

- Format all cells based on their values
- Format only cells that contain
- Format only top or bottom ranked values
- Format only values that are above or below average
- Format only unique or duplicate values
- Use a formula to determine which cells to format

Edit the Rule Description:

Format only cells with:

Specific Text containing B

Preview: AaBbCcYyZz

OK Cancel

Cut Copy Paste Format Painter Clipboard Font

11

Clipboard Font

J2

1 Roll no Name English Maths

2 20201-BCE-021 AHMED 55

3 20201-BCE-022 WAHAB 67

4 20201-BCE023 AYAN 31

5 20201-BCE-024 SARIM 20

6 20201-BCE-025 FATIMA 31

7 20201-BCE026 SARA 14

8 20201-BCE-027 ZARA 79

9 20201-BCE028 AYESHA 94

10 20201-BCE-029 SOHAIL 15

11 20201-BCE-030 ALI 88

12 Max. marks of subject 94

13 Min. marks of subjects 14

14

15

Format Cells

Number Font Border Fill

Background Color:

No Color

Pattern Color: Automatic

Pattern Style:

Fill Effects... More Colors...

Sample

Clear

OK Cancel

Clipboard Font Alignment Number Styles Cells

2

Clipboard Font Alignment Number Styles Cells

2

1 Roll no Name English Maths

2 20201-BCE-021 AHMED 55 44

3 20201-BCE-022 WAHAB 67 37

4 20201-BCE023 AYAN 31 52

5 20201-BCE-024 SARIM 20 31

6 20201-BCE-025 FATIMA 31 100

7 20201-BCE026 SARA 14 27

8 20201-BCE-027 ZARA 79 75

9 20201-BCE028 AYESHA 94 28

10 20201-BCE-029 SOHAIL 15 94

11 20201-BCE-030 ALI 88 43

12 Max. marks of subject 94 100 95 91 94

13 Min. marks of subjects 14 27 12 19 17

14

15

New Formatting Rule

Select a Rule Type:

- Format all cells based on their values
- Format only cells that contain
- Format only top or bottom ranked values
- Format only values that are above or below average
- Format only unique or duplicate values
- Use a formula to determine which cells to format

Edit the Rule Description:

Format only cells with:

Specific Text containing C

Preview: AaBbCcYyZz

OK Cancel

Format Painter

Clipboard Font

Formula Bar: `=IF(I2>=90, "A", IF(I2>=80, "B", IF(I2>=70, "C", "F")))`

New Formatting Rule

Select a Rule Type:

- Format all cells based on their values
- Format only cells that contain
- Format only top or bottom ranked values
- Format only values that are above or below average
- Format only unique or duplicate values
- Use a formula to determine which cells to format

Edit the Rule Description:

Format only cells with:

Specific Text containing F

Preview: No Format Set

OK Cancel

	A	B	C	D
1	Roll no	Name	English	Maths
2	20201-BCE-021	AHMED	55	44
3	20201-BCE-022	WAHAB	67	37
4	20201-BCE023	AYAN	31	52
5	20201-BCE-024	SARIM	20	31
6	20201-BCE-025	FATIMA	31	100
7	20201-BCE026	SARA	14	27
8	20201-BCE-027	ZARA	79	75
9	20201-BCE028	AYESHA	94	28
10	20201-BCE-029	SOHAIL	15	94
11	20201-BCE-030	ALI	88	43
12	Max. marks of subject		94	100
13	Min. marks of subjects		14	27

Stage	Grade
63	F
54	F
54.6	F
44.2	F
47.4	F
59.6	F
70.6	F
55	C
52.8	F
61.8	F

Format Cells

Number Font Border Fill

Background Color:

Pattern Color: Automatic

Pattern Style:

Sample

OK Cancel

	A	B	C	D
1	Roll no	Name	English	Maths
2	20201-BCE-021	AHMED	55	
3	20201-BCE-022	WAHAB	67	
4	20201-BCE023	AYAN	31	
5	20201-BCE-024	SARIM	20	
6	20201-BCE-025	FATIMA	31	
7	20201-BCE026	SARA	14	
8	20201-BCE-027	ZARA	79	
9	20201-BCE028	AYESHA	94	
10	20201-BCE-029	SOHAIL	15	
11	20201-BCE-030	ALI	88	
12	Max. marks of subject		94	
13	Min. marks of subjects		14	27

Format Cells

Number Font Border Fill

Background Color: AaBbCcYyZz

Pattern Color: Automatic

Pattern Style:

Sample

OK Cancel

	A	B	C	D
1	Roll no	Name	English	Maths
2	20201-BCE-021	AHMED	55	44
3	20201-BCE-022	WAHAB	67	37
4	20201-BCE023	AYAN	31	52
5	20201-BCE-024	SARIM	20	31
6	20201-BCE-025	FATIMA	31	100
7	20201-BCE026	SARA	14	27
8	20201-BCE-027	ZARA	79	75
9	20201-BCE028	AYESHA	94	28
10	20201-BCE-029	SOHAIL	15	94
11	20201-BCE-030	ALI	88	43
12	Max. marks of subject		94	100
13	Min. marks of subjects		14	27

Stage	Grade
63	F
54	F
54.6	F
44.2	F
47.4	F
59.6	F
70.6	F
55	C
52.8	F
61.8	F

	A	B	C	D	E	F	G	H	I	J
1	Roll no	Name	English	Maths	Physics	Chemistry	Computing	Total	Total Percentage	Grade
2	20201-BCE-021	AHMED	34	66	69	93	99	361	72.2	C
3	20201-BCE-022	WAHAB	2	72	11	58	3	146	29.2	C
4	20201-BCE023	AYAN	87	34	46	3	9	179	35.8	F
5	20201-BCE-024	SARIM	82	86	47	29	55	299	59.8	F
6	20201-BCE-025	FATIMA	50	70	96	74	37	327	65.4	F
7	20201-BCE026	SARA	83	5	14	58	95	255	51	F
8	20201-BCE-027	ZARA	48	72	4	98	42	264	52.8	F
9	20201-BCE028	AYESHA	93	14	12	42	58	219	43.8	F
10	20201-BCE-029	SOHAIL	68	37	8	88	59	260	52	F
11	20201-BCE-030	ALI	14	61	25	53	6	159	31.8	F
12	Max. marks of subject		93	86	96	98	99			
13	Min. marks of subjects		2	5	4	3	3			

11. Perform *VLOOKUP* on your data. Set the roll number as the lookup field. The data returned should be the percentage

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V
1	Roll no	Name	English	Maths	Physics	Chemistry	Computing	Total	Total Percentage	Grade												
2	20201-BCE-021	AHMED	70	13	23	80	66	252	50.4	F												
3	20201-BCE-022	WAHAB	71	80	98	75	53	377	75.4	F												
4	20201-BCE023	AYAN	22	37	6	48	77	190	38	C												
5	20201-BCE-024	SARIM	97	49	63	59	89	357	71.4	C												
6	20201-BCE-025	FATIMA	18	5	42	86	26	177	35.4	C												
7	20201-BCE026	SARA	75	96	25	49	20	265	53	F												
8	20201-BCE-027	ZARA	24	33	65	2	60	184	36.8	F												
9	20201-BCE028	AYESHA	58	88	22	70	5	243	48.6	F												
10	20201-BCE-029	SOHAIL	54	66	40	22	4	186	37.2	F												
11	20201-BCE-030	ALI	69	67	30	29	92	287	57.4	F												
12	Max. marks of subject		97	96	98	86	92															
13	Min. marks of subjects		18	5	6	2	4															
14																						
15																						
16																						
17																						
18																						
19																						
20																						
21																						

Roll no	Percentage
20201-BCE-021	=VLOOKUP(N11,\$A\$2:\$I\$11,9,0)
20201-BCE-022	VLOOKUP(lookup_value, table_array, col_index_num, [range_lookup])
20201-BCE023	38
20201-BCE-024	71.4
20201-BCE-025	35.4
20201-BCE026	53
20201-BCE-027	36.8
20201-BCE028	48.6
20201-BCE-029	37.2
20201-BCE-030	57.4

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P
1	Roll no	Name	English	Maths	Physics	Chemistry	Computing	Total	Total Percentage	Grade						
2	20201-BCE-021	AHMED	88	67	79	40	22	296	59.2	F						
3	20201-BCE-022	WAHAB	100	99	68	90	32	389	77.8	F						
4	20201-BCE023	AYAN	14	12	3	44	88	161	32.2	C						
5	20201-BCE-024	SARIM	25	59	69	42	66	261	52.2	F						
6	20201-BCE-025	FATIMA	13	29	47	19	13	121	24.2	F						
7	20201-BCE026	SARA	20	75	74	8	21	198	39.6	F						
8	20201-BCE-027	ZARA	51	91	52	56	14	264	52.8	F						
9	20201-BCE028	AYESHA	68	95	13	93	8	277	55.4	F						
10	20201-BCE-029	SOHAIL	16	98	71	37	53	275	55	F						
11	20201-BCE-030	ALI	13	74	72	53	27	239	47.8	F						
12	Max. marks of subject		100	99	79	93	88									
13	Min. marks of subjects		13	12	3	8	8									
14																
15																
16																
17																
18																
19																
20																
21																

Roll no	Percentage
20201-BCE-021	59.2
20201-BCE-022	77.8
20201-BCE023	32.2
20201-BCE-024	52.2
20201-BCE-025	24.2
20201-BCE026	39.6
20201-BCE-027	52.8
20201-BCE028	55.4
20201-BCE-029	55
20201-BCE-030	47.8

12. Design a TEMPERATURE CONVERTER from Celsius to Fahrenheit and Fahrenheit to Celsius. Present the Converter with appropriate design. Formulas for conversion are as follows:

$$[^{\circ}\text{C}] = ([^{\circ}\text{F}] - 32) \cdot \frac{5}{9}$$

$$[^{\circ}\text{F}] = [^{\circ}\text{C}] \cdot \frac{9}{5} + 32$$

Also report the weather conditions as HOT if the temperature exceeds 35°C, WARM if the temperature is between 20°C to 35°C and COLD if the entered temperature is below 20°C.

	A	B	C	D	E	F
1	TEMPERATURE IN (C OR F)	TEMPERATURE VALUE	IN FAHRENHEIT	IN CELSIUS	WEATHER CONDITION	
2						
3						
4						
5						
6						
7						
8						
9						
10						
11						

	A	B	C	D	E
1	TEMPERATURE IN (C OR F)	TEMPERATURE VALUE	IN FAHRENHEIT	IN CELSIUS	WEATHER CONDITION
2			=IF(A2="C",B2*9/5+32,B2)		
3					
4					
5					
6					
7					
8					
9					
10					

	A	B	C	D	E
1	TEMPERATURE IN (C OR F)	TEMPERATURE VALUE	IN FAHRENHEIT	IN CELSIUS	WEATHER CONDITION
2	C	75	167	=IF(A2="F", (B2-32)*9/5, B2)	
3					
4					
5					
6					
7					
8					
9					
10					

[illegible]

	A	B	C	D	E
1	TEMPERATURE IN (C OR F)	TEMPERATURE VALUE	IN FAHRENHEIT	IN CELSIUS	WEATHER CONDITION
2	C	46	114.8	46	HOT
3					
4					
5					
6					
7					
8					

	A	B	C	D	E
1	TEMPERATURE IN (C OR F)	TEMPERATURE VALUE	IN FAHRENHEIT	IN CELSIUS	WEATHER CONDITION
2	C	46	114.8	46	HOT
3	C	11	51.8	11	COLD
4	C	20	68	20	WARM
5					

13. Create a magic square puzzle as per given example below, the sum of all the numbers in a row must be equal, simultaneously the sum of all the numbers in a column must be equal, and the sum of diagonal numbers should also be equal:

- Take the input from the user in all the squares. For a user all the squares will be blank initially and the box given below will contain the text “KEEP TRYING” unless the user enters all correct entries. If user solves the puzzle correctly then a message “WELL DONE” appears below. Format your work accordingly.

The screenshot shows an Excel spreadsheet with the following details:

- Formula Bar:** Displays the formula `=SUM(D3:D5)`.
- Spreadsheet Grid:** Columns A through J and rows 1 through 13 are visible.
- Selection:** A dashed blue box highlights the range D3:D5.
- Cell D3:** Contains the formula `=SUM(D3:D5)`.
- Cell D4:** Contains the value 1.
- Cell D5:** Contains the value 2.
- Cell D6:** Contains the value 3.
- Cell D7:** Contains the value 4.
- Cell D8:** Contains the value 5.
- Cell D9:** Contains the value 6.
- Cell D10:** Contains the value 7.
- Cell D11:** Contains the value 8.
- Cell D12:** Contains the value 9.
- Cell D13:** Contains the value 10.

	A	B	C	D	E	F	G	H	
1									
2									
3				12	3	9		24	
4				5	8	11		24	
5				7	13	4		24	
6								24	
7			24	24	24	24			
8									
9			WELLDONE						
10									

	12	3	9	24
	5	8	11	24
	7	13	4	24
24	24	24	24	24
WELL DONE!!!				

NED University of Engineering & Technology
Department of Software Engineering

**Course Code & Title: CT-174 FIT Assessment Rubric
for Psychomotor (P2-PLO-3)**

Instructor: _____ Batch: 2024 Class: FESE Semester: Fall 2024.
Student's Name: _____ Roll No.: _____
Laboratory Session No. _____ Date: _____

Psychomotor Domain Assessment Rubric-Level P2-PLO3					
Skill Sets	Extent of Achievement				
	Below Average (0--1)	Average (2)	Good (3)	Very Good (4)	Excellent (5)
Problem Definition and Understanding	Unable to define and understand the problem clearly.	Struggles to define the problem clearly, with significant gaps in understanding	Defines the problem, but with gaps or a limited understanding of key components, impacting the overall quality of definition.	Effectively defines the problem with a good understanding of its components, showcasing proficiency in problem definition.	Demonstrates an exceptional understanding of the problem, identifying all key components and challenges with insightful depth.
Technical Implementation <i>Proficient use of tools to complete the problem</i>	Struggles to use tools effectively, requiring significant improvement.	Uses tools with noticeable inefficiencies.	Demonstrates basic proficiency in utilizing tool, with some room for improvement.	Effectively uses tools to support the task.	Integrates use of tools seamlessly to enhance the design and development process
Solution Design& Accuracy	Struggles to relate the implemented solutions to the design and development goals, requiring significant improvement.	Shows limited connection between the implemented solutions and the design and development context.	Demonstrates a basic understanding of how the solutions align with design and development requirements.	Effectively relates the implemented solutions to the overall design and development objectives	Maps the implemented solutions seamlessly to the overarching design and development goals, demonstrating a comprehensive understanding.
Trouble Shooting Skills Issue resolution skills	Unable to trouble shoot	Limited troubleshooting	Adequate troubleshooting, minor errors remaining	Good troubleshooting, no errors remaining	Excellent trouble shooting no errors remain without delay.
Document Organization and on time submission	Poor Unclear disorganized document late submission	Somewhat clear with lack of organization and structuring late submission	Somewhat clear with lack of organization and structuring on time submission	Reasonably clear and organized with adequate structuring on time submission	Highly clear and well-structured document, On time submit

Weighted CLO (Psychomotor Score)		
: Instructor's Signature with Date		